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A linked carbon cycle and crop developmental model: Description and evaluation against measurements of carbon fluxes and carbon stocks at several European agricultural sites

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ABSTRACT

Croplands are the largest biospheric source of carbon (C) to the atmosphere in Europe each year (~0.3 GtC y^{-1}). Moreover, the biological dynamics of managed landscapes affect the fluctuations of atmospheric CO_2 levels on annual and interannual scales. The activity of croplands must therefore be included in efforts to quantify, understand and regulate the global C cycle. C flux data for croplands, existing mostly from ~2003 onwards, provide an opportunity to better understand the timing and magnitude of C exchanges in croplands and can help to improve crop C modelling on various spatiotemporal scales. A crucial process to be considered for C budget modelling is crop development, to which the C allocation pattern and senescence and remobilisation processes are directly linked.

We developed, parameterised, and tested a new crop C mass balance model (SPAc), which included all major developmental stages (DS) and C remobilisation. Model results were compared against independent observations of C fluxes and biometric data for eight site years, at six European sites growing winter wheat and barley. We used a single parameterisation for all sites, varying only maximal carboxylation rate (V_{cmax}), maximal photosynthetic electron transport rate (J_{max}), and C per leaf area (C_{la}) between wheat and barley. The model effectively described the seasonal dynamics of net ecosystem exchange (NEE) through simulation of DS. We found high model reliability for predicting daily NEE fluxes (mean R^2 = 0.83, mean RMSE = 1.47 gC m^{-2} d^{-1}). Results for cumulative NEE from sowing to harvest were of lower quality (mean R^2 = 0.27). The model tended to overpredict leaf area index (LAI, 160% of observed), and underpredict final yield (71% of observed). The introduction of a dead leaf C pool considerably improved at-harvest estimates of total leaf C mass. Simulating the shift from green to brown above-ground biomass is of potential significance for solving the surface energy balance and for future remote sensing studies through an improved simulation of crop canopy optical properties.

1. Introduction

In 2000, about 12% of the Earth's ice-free land surface was cropland, with pasture accounting for a further 22% (Ramankutty et al., 2008). These managed lands play a significant role in global biogeochemical cycles (Denman et al., 2007). The global human appropriation of net primary production (the aggregate impact of land use on the biomass produced each year in ecosystems) has been estimated to amount to ~25% (Haberl et al., 2007), with a significant share of this ascribed to biomass removal during harvest (~53%). Janssens et al. (2003) estimated croplands to be the largest biospheric source of C to the atmosphere in Europe each year (~0.3 GtC y^{-1}, for Europe as far east as the Urals). The biological dynamics of managed landscapes affect the fluctuations of atmospheric CO_2 levels on annual and interannual scales (Moureaux et al., 2008; Houghton, 1999). The activity of croplands must therefore be included in efforts to quantify, understand and regulate the global C cycle, for example through the United Nations Framework Convention on Climate Change.

Currently there is limited understanding of the interplay between land management and C cycling, and more specifically between management practices, agricultural yield and net C bal-

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ance (Smith et al., 2010). Studies on cropland C balance, linking both plant and soil C dynamics, have been rare. Agronomists have largely focused on measuring and modelling crop yields and their dependence on crop management and intervention, rather than the seasonal cycle of C fluxes and how these differ from unmanaged ("natural") systems. The observation, analysis and modelling of C fluxes since the 1990s, when it began in earnest, has largely focused on these unmanaged ecosystems. To date, only 30 out of the 251 sites contained within the FLUXNET La-Thuile data set are croplands (Williams et al., 2009).

C flux data for croplands exist mostly from ∼2003 onwards (with a few earlier exceptions, e.g. the Ameriflux network provides cropland NEE data from 1997 onwards for their Bondville site). These data provide an opportunity to better understand the timing and magnitude of C exchanges in croplands and can help to improve crop C modelling on various spatiotemporal scales. However, land surface models have mostly lacked crop-related plant functional types (PFT), and often the grassland PFT has been used as a proxy for crops (Osborne et al., 2007). But now crop C budget modelling studies are taking into account eddy flux information (Adiku et al., 2006; Arora, 2003; Huang et al., 2009; Lokupitiya et al., 2009; Wang et al., 2005, 2007), and global land surface models, which incorporate land management, are being developed and tested—such as LPJmL (Bondeau et al., 2007), Agro-IBIS (Kucharik and Twine, 2007) and ORCHIDEE-STICS (de Noblet-Ducoudré et al., 2004). One of the major challenges in modelling crops is to simulate the complete spectrum of plant development, from seed to senescence. This spectrum is important because developmental stages represent shifts in C allocation and remobilisation patterns. Natural vegetation C models tend not to include such shifts, so significant changes to C models are required.

Both a realistic representation of the aforementioned crop developmental pattern and the consideration of soil C dynamics are required in order to effectively simulate C dynamics in croplands. Whilst physiological models exist for specific crops or cropping systems (Jamieson et al., 1998; Keating et al., 2003; Porter et al., 1983; Stockle et al., 1994; Van Laar et al., 1997; Williams et al., 1989; Yang et al., 2004), no existing Dynamic Global Vegetation Model (DGVM) includes, for example, plant development, C remobilisation, and senescence, all within a full crop C budget. For all but one (Wang et al., 2007) of the models mentioned above, crop development is based on a linear relationship to cumulative temperature (growing degree days, GDD), photoperiod, and vernalization. Furthermore, the models do not account for C remobilisation in the reproductive phase, nor do they contain a standing dead leaf C pool. Additionally, in some of the models, the evolution of potential leaf area index (LAI) is either prescribed (Bondeau et al., 2007; de Noblet-Ducoudré et al., 2004) or constrained through a maximum LAI value (Kucharik and Twine, 2007; Huang et al., 2009), or leaf senescence rate was set to a constant value (Adiku et al., 2006).

In this paper we present a new model, which simulates the full crop C balance, and explicitly models all major developmental stages, including senescence, with justifiable non-linear relationships to temperature, photoperiod and vernalisation, as well as C remobilisation. Further, we evaluate the model outputs against independent observations and thus determine model validity and weaknesses. Our objective here was to compare model results with both CO_2 flux data and time series of biomass harvests from a selection of six European cropland sites over multiple years. Out of the crop C budget models mentioned above, only SiBcrop (three sites, Lokupitiya et al., 2009), LPJmL (three sites), and ORCHIDEE-STICS (two sites) have been compared to C flux data measured at several sites. We aimed to determine (1) how effectively the model described observed fluxes and C biomass changes as the crops developed, and (2) critical weaknesses in model formulation and parameterisations. One major advantage of using eddy covariance data is their temporal resolution, which allows predicted C exchanges to be evaluated over a range of time scales, weather conditions, and, particularly important in this case, crop developmental stages.

Our further objectives were to determine whether different crop types and European locations could be accommodated within a single model structure, with resulting predictions of C fluxes and biomass that differ from observations only within acceptable limits. We also undertook a detailed sensitivity analysis to identify which parameters controlled NEE and model error.

This study is novel in two ways. First, the model includes a broadly applicable and advanced representation of crop development based on Penning de Vries et al. (1989), Wang and Engel (1998), and Streck et al. (2003), within a full C budget framework. Secondly, we have evaluated the model against multiple data sets with an explicit focus on parameter sensitivity. Hence we present our insights into both model weaknesses and requirements for field studies that can constrain critical parameters.

2. Data and methods

2.1. Data and study sites

We compared modelled estimates of NEE against observations of NEE made over croplands by the eddy covariance method, derived from the CarboEurope database (http://gaia.agraria.unitus.it/database/carboeuropeip/). We used gap-filled half-hourly level 4 data (Falge et al., 2001; Reichstein et al., 2005), so that an observation value existed for each model time-step (0.5 h) and each simulation day. The gap-filling procedure introduced empirically modelled estimates into the gaps within the data record. Consequently, the process modelling was compared to a mix of observations and empirical interpolations. As daily NEE values are aggregates of half-hourly observations, the percentage of gap-filled values used for estimating daily NEE was variable. Around 30–60% of half-hourly NEE values within a growing seasons overall record were flagged as gap-filled (Table 1). However, for each of the NEE data time series used in this study, the percentage of half-hourly values falling in the quality classes 0 (original value) or 1 (most reliable gap-filled) is >90%.

We selected six sites from the CarboEurope database (Fig. 1), and data collected over the years 2005–2007. The sites are located in France (Auradé + Lamasquère, Béziat et al., in press; Grignon), Germany (Gebesee, Anthoni et al., 2004; Klingenberg) and Switzerland (Oensingen, Dietiker et al., 2010). They vary in latitude (from 43.6°N to 51.1°N) and longitude (from 1.1°E to 13.5°E), and represent considerable variety in terms of western-central European climate. Average temperatures measured during the various growing periods (sowing–harvest) ranged from 6.0 to 11.3, and precipitation from 327 to 1051 mm. These climates result in a substantial variation in growing periods over the six sites. Particularly high average temperatures and precipitation totals have been recorded for the growing seasons of those crops planted in late 2006 and harvested in the summer of 2007. The amount of precipitation measured during growing season Oensingen(b) is outstanding with a reported value of ∼1050 mm in 270 days. Growing periods not only varied with respect to timing (with sowing dates ranging from mid September to early November, i.e. a difference of 53 days maximum) but also overall length (from 245 to 347 days, Table 1). Field measurements of soil texture values indicate varying local pedo-climatic conditions, although loamy soils predominate.

Overall, the selection of study sites and years offers a broad spectrum of western-central European climatic conditions, which were used to test model performance. Using these data, we compared simulated and observed NEE for winter wheat (ww) and winter

Table 1
List of study sites showing length of crop growing period (from sowing to harvest), the percentage of half-hourly NEE data that were gap-filled, average temperature (av. temp.) and precipitation (precip.) during the growing period, and soil texture (Clay:Sand:Silt in percent) and type. In column 1, winter barley sites are marked (*), all others are winter wheat.

Site	Growing period	Length [days]	Gap-filled NEE [%]	Av. temp. [°C]	Precip. [mm]	Soil texture (Cl:Sa:Si [%]) and type
Auradé	27.10.05 to 29.06.06	245	43.9	9.7	374	32.3:20.6:47.1 luvisol
Gebesee(a)*	17.09.04 to 16.07.05	302	29.2	7.9	366	35.8:3.9:60.3 chernozem
Gebesee(b)	09.11.06 to 07.08.07	271	45.9	10.6	447	
Grignon	28.10.05 to 14.07.06	259	40.3	8.2	327	18.9:7.0:74.1 luvisol
Klingenberg	25.09.05 to 06.09.06	347	42.2	6.0	607	44.1:21.7:34.24 pseudogley
Lamasquère	18.10.06 to 15.07.07	270	52.4	11.3	531	54.3:12.0:33.7 luvisol on alluvium
Oensingen(a)*	29.09.04 to 07.07.05	281	46.6	7.5	548	43.0:9.5:47.54 eutric-stagnic cambisol
Oensingen(b)	19.10.06 to 16.07.07	270	52.5	10.2	1051	

barley (wb) over 6 (ww) and 2 (wb) growing periods. Wheat and barley are the two main cereals (including grain maize but not green or silage maize) in Europe (EU-27), with a harvested production in 2007 of 46% (wheat) and 22% (barley) of the total for cereals (~260 million tonnes) Eurostat, 2008.

2.2. Model description

2.2.1. Photosynthesis, energy and water balance

The Soil Plant Atmosphere (SPA) model (Williams et al., 1996, 2001b) is a process-based model that simulates ecosystem photosynthesis and water balance at fine temporal and spatial scales (30 min time-step, up to ten canopy and twenty soil layers). The scales of parameterization (leaf-level) and prediction (canopy-level) have been designed to allow the model to diagnose eddy flux data and to provide a tool for scaling up leaf-level processes to canopy and landscape scales (Williams et al., 2001b). The SPA model employs some well-tested theoretical representations of ecophysiological processes, such as the Farquhar model of leaf-level photosynthesis (Farquhar and von Caemmerer, 1982), and the Penman-Monteith equation to determine leaf-level transpiration (Jones, 1992). These two processes are linked by a model of stomatal conductance that optimizes daily C gain per unit leaf nitrogen, within the limitations of canopy water storage and soil to canopy water transport. Stomata adjust in order to maximize C assimilation within the limitations of the hydraulic system, so stomatal resistance is adjusted to balance atmospheric demand for water with rates of water uptake and supply from soils. As a consequence, evapotranspiration is maintained at a rate that keeps leaf water potential [Ψ_1] from falling below a critical threshold value, below which potentially dangerous cavitation of the hydraulic system may occur (Williams et al., 1996).

Plant hydraulics within SPA are represented as the change of leaf water potential over time. This change is a function of soil water potential, capacitance of the hydraulic system, evapotranspiration, and above and below ground hydraulic resistance. Based on these assumptions and the biochemical parameters for maximum carboxylation capacity (V_{cmax}) and maximum electron transport rate

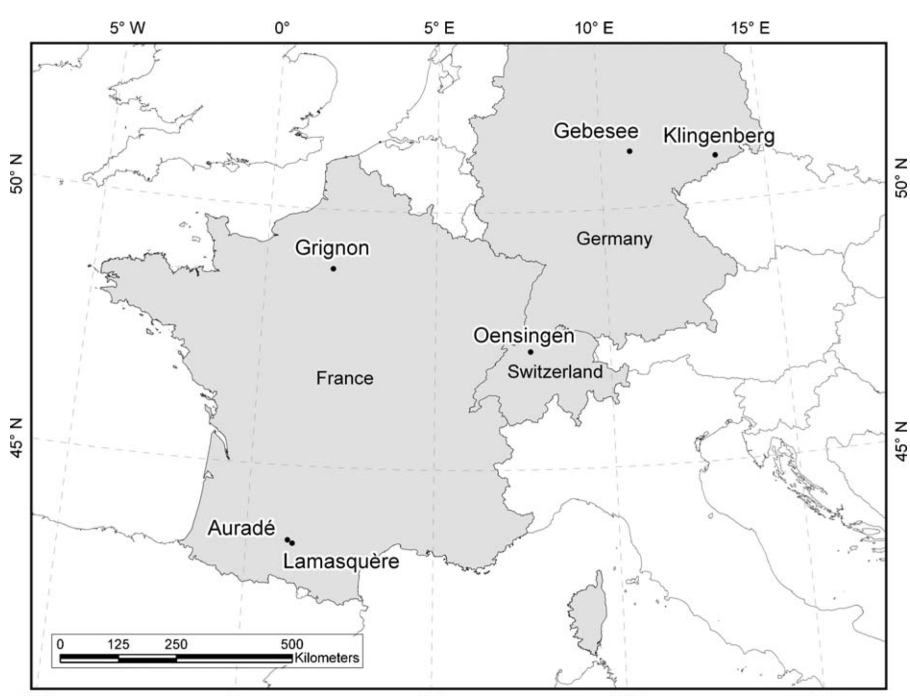

Fig. 1. Locations of study sites.

(J_{max}), for each canopy layer and time-step an iterative procedure is used to determine the maximum sustainable stomatal conductance (g_s) and the assimilation rate associated with this conductance. SPA also contains a detailed representation of soil hydrology and thermal dynamics, allowing the surface energy balance to be solved as a function of down-welling radiation reaching the soil surface by estimating the soil surface temperature. Net radiation is then partitioned into sensible, latent and ground heat fluxes (Williams et al., 2001a). For each soil layer, changes to water content are regulated by precipitation and evaporation (surface layer only), abstraction by roots (rooted layers only), and gravitational drainage, whilst soil porosity and water retention curves of each layer are estimated according to empirical relationships with soil texture. Plant root distribution is determined by total fine root biomass, maximum root biomass per unit volume, and depth of rooting. LAI is calculated by dividing live leaf C mass by the C per leaf area parameter C_{la}. The SPA model has been applied in natural ecosystems ranging from 70°N to 2°S (Williams et al., 1989, 2000). More recently, a C mass balance model has been developed (Williams et al., 2005) to link to SPA.

Below, we describe modifications that have been made in order to simulate the dynamics of agroecosystems. These modifications are (1) the addition of a new developmental module dependant on ambient temperature, day length, and vernalization and (2) the modification of the C allocation pattern in terms of adding/removing C pools and introducing a dependence of C allocation on crop developmental stage.

2.2.2. Carbon partitioning scheme

The C partitioning scheme embedded within SPA version 2 – Crop (hereafter referred to as SPAc; SPA as modified for the simulation of crop C dynamics) is based on observations of field crops with a series of harvests and crop growth analyses (Penning de Vries et al., 1989). Time series of C allocation fractions to the various crop plant organs (root, leaves, stem, storage organs) are derived from the patterns of increase in biomass. These fractions account for the efficiency of glucose conversion into structural dry matter. We chose this crop C partitioning scheme as corresponding tables have been published in Penning de Vries et al. (1989) for 14 different annual crop types such as wheat, barley, maize, potato, rice, sugar-beet, and soybean among others. As a consequence, information on C partitioning for wheat and barley were readily available for this study and new crop functional types can easily be added to SPAc. The original range of developmental stages, as can be found in Penning de Vries et al. (1989), has been extended to account for pre-emergence development as proposed by Wang and Engel (1998).

2.2.3. Crop developmental model

In the scheme outlined above, C allocation keys are given as a function of developmental stage (DS), which quantifies a crop plant's physiological age, and is related to its morphological appearance (for an example see Fig. 2). It is important to correctly simulate crop plant development, as the C allocation pattern is directly related to it (Penning de Vries et al., 1989). The model representation of crop developmental stage is introduced into SPAc by a new state variable, ranging from −1 at sowing to 2 at maturity (see Table 2 for intermediate phases and stages). DS is calculated as the sum of daily values of developmental rate (DR).

$$DS = \sum DR \quad [\mathrm{d}^{-1}] \tag{1}$$

A modified Wang & Engel model (Streck et al., 2003) was implemented in SPAc to calculate crop developmental rate on the basis of justifiable non-linear functions for three environmental factors: temperature $f(T)$, photoperiod $f(P)$, and vernalization $f(V)$. Only severe water stress is known to have a direct influence on crop development (Penning de Vries et al., 1989), and so soil moisture effects are not considered here.

Table 2
Relation of SPAc developmental stages (DS) to crop growing phases and stages. DS [d^{-1}] ranges between −1 at sowing and +2 at maturity. After: Wang and Engel (1998).

Phase	Start of stage	DS
Pre-emergence	Sowing	−1
	Germination	−0.5
Pre-anthesis (vegetative)	Emergence	0
	Floral initiation	0.2
	Terminal spikelet	0.45
	Flag leaf	0.65
	Heading	0.9
Post-anthesis (reproductive)	Anthesis	1.
	Milk development	1.15
	Dough development	1.5
Ripening	Ripening	1.95
	Maturity	2.

In detail, DR is calculated as the product of a maximum developmental rate DR (the maximum possible DR under optimal ambient conditions for development) and the three developmental response functions.

$$DR = DR_{max} \times f(T) \times f(P) \times f(V) \quad [\mathrm{d}^{-1}] \tag{2}$$

The temperature response function is:

$$f(T) = \frac{[2(T - T_{min})^{\alpha} \times (T_{opt} - T_{min})^{\alpha} - (T - T_{min})^{2\alpha}]}{(T_{opt} - T_{min})^{2\alpha}} \tag{3}$$

for $T_{min} \le T \le T_{max}$. $f(T) = 0$ for $T < T_{min}$ or $T > T_{max}$. The exponent α is given by:

$$\alpha = \frac{\ln 2}{\ln[(T_{max} - T_{min})/(T_{opt} - T_{min})]} \tag{4}$$

where T_{min}, T_{opt}, and T_{max} are the cardinal temperatures for development (minimum, optimum, and maximum), and T is the mean daily temperature calculated from the 48 half-hourly temperature values.

The photoperiod response function is:

$$f(P) = 1 - \exp[-\omega(P - P_c)] \tag{5}$$

where P is the actual photoperiod, P_c the critical photoperiod below which no development occurs, and ω is a cultivar-specific photoperiod sensitivity coefficient.

The effective vernalization days, VD, are calculated from sowing as:

$$VD = \sum fvn(T) \tag{6}$$

where $fvn(T)$ is the daily vernalization rate (per day), calculated using Eqs. (3) and (4) with the cardinal temperatures for vernalization ($T_{min,v}$, $T_{opt,v}$, and $T_{max,v}$). The vernalization response function is:

$$f(V) = \frac{(VD)^5}{(VD_h)^5 + (VD)^5} \tag{7}$$

where $f(V)$ is the vernalization function that varies from 0 (unvernalized) to 1 (fully vernalized plants). Parameter VD_h is the effective vernalization days when plants are 50% vernalized, and the exponent 5 provides the sigmoidal shape of the response to VD. We understand vernalization as "the acquisition or acceleration of the ability to flower by a chilling treatment" (Chouard, 1960). According to Slafer and Rawson (1994), it is generally accepted that exposure to vernalising temperatures after seed imbibition (i.e. absorption of water by seed) can affect wheat developmental rate during the

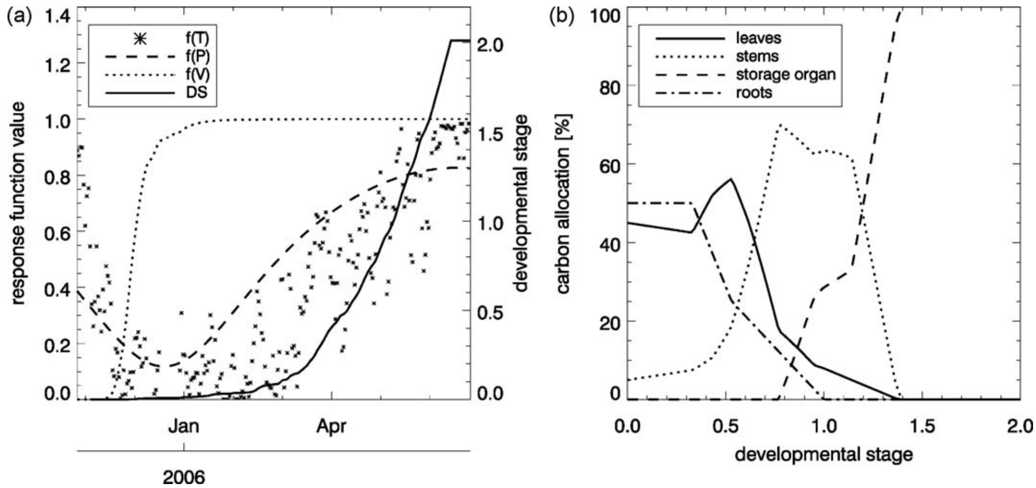

Fig. 2. (a) Time vs. developmental response function values ($f(T)$ = temperature function, $f(P)$ = photoperiod function, $f(V)$ = vernalization function) plus developmental stage (DS) and (b) DS vs. carbon allocation pattern for growing season Aurad (winter wheat) as implemented in SPAc. In order to maximise plotting space, DS values <0 (sowing to emergence) are not shown in (a) and (b).

vegetative stages. For a more detailed description of these functions see Streck et al. (2003).

In SPAc, $f(P)$ and $f(V)$ delay development only in the vegetative phase ($f(T)$ in the vegetative and reproductive phases), and different DR_{max} are assumed for the vegetative and reproductive phases (Streck et al., 2003). In our experiments, developmental rate is initially limited by $f(V)$ values. However, as $f(V)$ saturates quickly through the winter period ($f(V) = 1$), $f(P)$ and $f(T)$ become the only limiting factors on crop development during the vegetative phase (Fig. 2(a)). $f(T)$ remains limiting until crop maturity, but exhibits a continuously weakening influence ($f(T)$ close to 1) as mean daily temperatures increase throughout spring towards summer. As a result, DS remains low until the beginning of February, after which it rises gradually. By mid-March, developmental rate increases significantly so that DS rises in an quasi-exponential fashion until crop maturity. Due to space restrictions, Fig. 2(a) shows the progression of DS for Auradé only. However, all the other sites show a relatively similar phenological pattern, and all crop plants reached maturity (DS = 2) before harvest.

In the study of Streck et al. (2003), the modified Wang & Engel model produced a root mean squared error (RMSE) of simulated winter wheat development which was up to 45% lower compared to the original model. The only difference between the modified Wang & Engel model (Streck et al., 2003) and the way it is being used in SPAc is that we do not split the vegetative phase into two sub-phases. In the modified Wang & Engel model, $f(V)$, $f(P)$, and $f(T)$ are used in sub-phase I ($0 < DS < 0.4$), and $f(V)$ is deactivated in sub-phase II ($0.4 < DS < 1$). In SPAc, $f(V)$, $f(P)$, and $f(T)$ affect developmental rate throughout the vegetative phase (DS < 1), but only $f(T)$ is used in the reproductive phase (DS > 1). Another advantage of the modified Wang & Engel model is that both the temperature and vernalization response functions can easily be modified by changing their cardinal temperatures in order to improve representativeness for different developmental stages, agricultural regions and crop types or cultivars if required (though the vernalization function was found to be cultivar independent; Streck et al., 2003). Similarly, the photoperiod response function can be adapted by changing either the photoperiod sensitivity coefficient or the critical photoperiod.

2.2.4. Carbon fluxes: growth, respiration, senescence, and remobilisation

Based on this development-linked C allocation pattern, SPAc simulates the allocation of carbohydrates to one root and four above ground (i.e. shoot) C pools (labile, foliage, stem and storage organ C pools, Fig. 3). Around sowing (DS < 0), C gained through photosynthesis (after having subtracted autotrophic respiration as a fixed fraction (f_a) of gross primary productivity (GPP), Waring et al., 1998; Gifford, 2003) is mainly allocated to leaves and roots at approximately equal amounts (Fig. 2(b)). In the case of winter cereals, the root share decreases continuously from DS ≈ 0.3 onwards, leaf allocation considerably drops from DS ≈ 0.5, whilst stem allocation reaches maximum values ($0.7 < DS < 1.2$). For DS > 1.2, allocation to the storage organs (here cereal grains) increases sharply with declining partitioning to the stem C pool. All available C will be allocated entirely to the storage organ pool for DS > 1.4.

A litter C pool receives litter fluxes originating from the shoot and root C pools (Fig. 3). The root litter flux is a function of developmental stage and commences at flowering (DS = 1). Shoot litter fluxes only occur at harvest and are dependent on the amount of above-ground biomass left on field after harvest. Decomposition fluxes are then calculated as a function of litter C content, specific turnover rate parameters and temperature (based on a Q_{10} temperature relationship). In the same fashion, mineralization fluxes of the litter and soil organic matter C pools (i.e. heterotrophic respiration) are calculated (Williams et al., 2005). C originating from the fraction of GPP to be respired enters a respiratory C pool. The autotrophic respiration flux itself originates from the gradual turnover of this pool, and thus occurs throughout day and night, but is not directly determined by temperature.

The model runs were initiated at the reported sowing date, when, based on thermal time, SPAc calculates the duration of the period from sowing to emergence (a phase of non-activity). In the current version, SPAc does not explicitly account for germination and growth of roots and shoots from germination until emergence. At emergence, the gradual turnover of the labile C pool content (representing the amount of C contained in the seeds) is the only initial source available for growth. As leaves grow, photosynthesis initiates and new C is available for allocation. In SPAc, seedlings do not fully exhaust their seeds, as photosynthesis takes over the C supply once C available from photosynthesis is greater than C available from seed reallocation (Penning de Vries et al., 1989). The remaining labile (or seed) C will be turned over and added to the litter C pool. Photosynthesis continues throughout the late autumn and winter months at a reduced rate, only limited by local meteorology (i.e. no additional constraint on growth was introduced to represent the effect of "dormancy"), and will increase considerably with rising ambient temperature and global irradiation levels in spring.

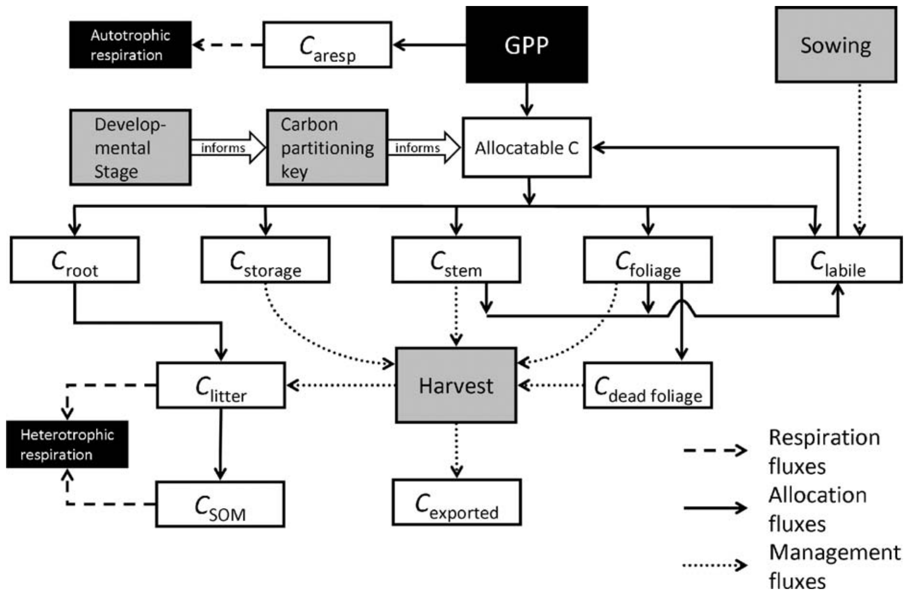

Fig. 3. Schematic of SPAc respiratory processes and photosynthesis (black boxes), carbon pools (white boxes), fluxes (arrows), and external drivers (grey boxes).

Later, in the reproductive phase, carbohydrates from the foliar and stem C pools are remobilized and transferred to the labile C pool. Remobilization of stem C is triggered once the running mean of the storage organ growth rate drops below a previously attained value. Leaf senescence rate on the other hand occurs either due to self-shading as a function of leaf area (if LAI is higher than the critical LAI threshold value) or ageing as a function primarily of developmental stage, whichever is higher (Van Laar et al., 1997). 50% of the senescing leaf C flux is treated as remobilized carbohydrate (Penning de Vries et al., 1989; Van Laar et al., 1997), whilst the other half enters a dead foliage C pool (Fig. 3). We assume that wheat/barley plants do not shed their leaves before harvest. Root death commences at flowering with a minimal value and linearly approaches and maintains its maximum rate at developmental stage 1.3 and beyond. In this simulation, dying roots are assumed not to contribute remobilised C to the labile C pool. Each of the remobilization processes involves a fixed respiratory cost due to the conversion between starch and glucose (Goudriaan and Van Laar, 1994). In the reproductive phase, young storage organs may not have the capacity to accept all the carbohydrates provided (allocated NPP plus remobilized reserves from stems and leaves), as there may be a high total number of receiving growing points, which still might have too small an overall sink size, though. Consequently, the growth rate of the storage organ is limited by a maximum potential growth rate per day, which is set to about a third of the overall storage organ pool (Goudriaan and Van Laar, 1994). If this maximum growth rate is exceeded, any surplus C will be allocated to the labile C pool.

At harvest (date as reported in Table 1), the fraction of the aboveground C mass exported from the field is given by the storage organ content plus a reported fraction of harvested stem and leaf C, and the residual crop C mass enters the litter C pool. The longer term fate of crop residues depends on the local residues management such as ploughing frequency or straw incorporation. Accordingly, crop residue decomposition and mineralization rates are affected by management type and timing, which have to be accounted for when simulating post-harvest ecosystem respiration fluxes. The model results presented here are only truly representative of the growing period between sowing and harvest, as ploughing, post-harvest sowing, and regrowth of volunteers are not explicitly considered.

2.2.5. Model setup

The input drivers for the SPAc model are time series of gap-filled half-hourly observations of temperature, ambient CO_2, wind speed, global radiation, vapour pressure deficit, precipitation and air pressure. Moreover, soil texture is prescribed as reported from the individual flux sites, and initial soil organic matter as well as labile C contents (i.e. seed C content) are estimated based on field observations reported in the literature (Anthoni et al., 2004; Aubinet et al., 2009; Halley and Soffe, 1988). Accordingly, initial values of soil organic matter in the top 40 cm soil layer and labile C content have been set to 7200 and 9 $\mathrm{gC\,m^{-2}}$, respectively. SPAc has been set up to include 4 canopy and 20 soil layers. The runs were initiated at sowing (date as reported in Table 1) and terminated by the end of the following year, even if cover crops were grown (e.g. Oensingen(a)). The effects of fertilization have not been taken into account, but reported harvest dates and crop residue management are considered in the model runs.

2.3. Sensitivity analysis and parameterisation

An analysis of model output sensitivity to a set of parameters has been performed in order to rate parameters by their importance in determining model behaviour. As a result, we are able to evaluate model reliability in more detail and recommend further needs of research on specific model processes and related parameters.

A single parameter sensitivity index (S) has been applied in this study. This index quantifies model sensitivity by relating the relative change of the state variable of interest with the relative change of the model parameter of interest. S is given by

$$S = \frac{(R_a - R_n)/R_n}{(P_a - P_n)/P_n} \quad (8)$$

where R_a and R_n are responses for altered and nominal model state variable(s) or statistics, and P_a and P_n are the altered and nominal parameters, respectively. We analysed model sensitivity for cumulative NEE (sowing to harvest) for the winter wheat growing period of Auradé. The relative changes to the diagnostics have been quantified for a set of 32 model parameters used within various SPAc modules, such as those simulating photosynthesis, crop development, plant hydraulics, and ecosystem respiration. For each sensitivity test, the particular parameter of interest was altered by adding (subtracting) 25% to (from) its nominal value, so that the denominator of equation (8) is always equal to either +0.25 (parameter increased by 25%) or −0.25 (parameter decreased by 25%). Consequently, if $S = 1$, a 25% increase (decrease) of a parameter leads to a 25% increase (decrease) of cumulative NEE. The following parameters are altered in a different fashion: only the absolute value of minimum leaf water potential Ψ_1 is modified (as its nominal value is already negative), leaf nitrogen distribution through the canopy layers is compared to a nominal distribution, only the decimal places of stomatal efficiency are altered, and each of the cardinal temperatures was varied by ±2.

An overview of model parameters, nominal values, and references is given in Table 3. The parameterisation is based on an extensive literature review, ensuring that all of our parameter values range within realistic, previously reported limits. We decided to use this single parameterisation (except for V_{cmax}, J_{max}, and C_{la}, which are crop type specific here) for each of the sites included in this study in order to examine whether it is possible to find a generic parameterisation that allows for a realistic field-scale simulation of winter wheat/barley C budgets over western-central Europe, and to highlight which parameters might need recalibration with varying latitudes and longitudes.

3. Results

3.1. Results for modelled carbon exchange data

3.1.1. Daily NEE flux data and residuals

A comparison of observed with modelled NEE flux data showed that SPAc modelled both the overall magnitude, and especially the seasonality, of C exchange for all of the 8 growing periods with high accuracy (mean $R^2 = 0.83$ (linear Pearson correlation coefficient), mean RMSE = 1.47 gC m^{-2} d^{-1}, Fig. 4). In general, a rather gradual decrease of observed NEE from sowing onwards (i.e. a gradual strengthening of the ecosystem's C uptake potential) until early spring compared well to modelled values. By the beginning of March, the assimilation of C increased considerably until reaching its peak value in a period between May and June. The observations were largely reproduced by the simulated NEE values, with SPAc results matching observations best for Auradé (ww), Gebesee(a) (wb), and Grignon (ww). Note that in Fig. 4(a), two modelled lines are shown. The line with higher post-harvest NEE displays results for all modelled above-ground C (except for grains) remaining on field after harvest, whereas the line with lower post-harvest NEE represents model results for 90% of both leaf and stem C having been exported. As a consequence, post-harvest decomposition and mineralisation (and thus NEE) fluxes are considerably different between the two. The highest observed daily rate of C assimilation was rather similar for all sites and ranged in between −8 and −12 gC m^{-2} d^{-1}, except for Oensingen(b) (ww), which had some days with productivity in excess of these values. Here, the observed peak value of C assimilation was ∼−15 gC m^{-2} d^{-1}, and the corresponding modelled value was considerably smaller in magnitude (∼−8 gC m^{-2} d^{-1}). After this phase of peak C assimilation, the observations showed that NEE became less negative (i.e. the ecosystems assimilatory capacity decreased) in a very rapid fashion towards crop maturity and harvest. The data indicated that all of the studied ecosystems were net sources of C by harvest, and this shift in C flows was well simulated.

There was a mismatch in terms of the timing of the onset of spring C assimilation for Klingenberg (ww). Modelled NEE decreased considerably around early March, whereas observations implied an onset of the peak growing season approximately a month and a half later.

At harvest, the reported fraction of crop residue and harvested biomass determined the allocation of above-ground C mass to either the litter C pool or the exported C pool. Consequently, modelled heterotrophic respiration increased at harvest proportionally to the amount of crop residue instantly available for decomposition and mineralization. Observed harvest-time NEE fluxes compared well to the simulated values, with exceptions: growing seasons Gebesee(a) (wb), Grignon (ww), and Klingenberg (ww) showed an increase in NEE around harvest date, whereas a much smaller (if any) influence of harvest on NEE could be seen for Gebesee(b) (ww), and Lamasquère (ww). Also a comparison of post-harvest values showed that for Lamasquère (ww) there was a relatively small difference in modelled and observed NEE fluxes, whilst this difference was considerably higher for Gebesee(b) (ww) and especially Oensingen(a) (wb), where a cover crop was sown after the wb harvest. There was a clear sensitivity in the model to assumptions about residues. Post-harvest decomposition and mineralisation (and thus NEE) fluxes were considerably different depending on whether all above-ground C (except for grains) was left on the field after harvest, or whether 90% of both leaf and stem C were exported (Fig. 4(a)).

For simplicity, we are not showing a comparison of modelling results of GPP and ecosystem respiration with values derived from NEE observations. However, these data are analysed in a companion paper in this special issue for growing periods Auradé, Gebesee(b), Grignon, Klingenberg, and Oensingen(b). Mean values of the Kendall tau rank correlation coefficient (r, modelled vs. observed values) for these sites are 0.65 (GPP) and 0.63 (ecosystem respiration). For a detailed analysis of these data please see Wattenbach et al. (2010).

Residuals of NEE (i.e. observed minus modelled NEE) indicated some clear periods of autocorrelation, associated with either crop development or post-harvest periods (Fig. 5). The large post-harvest residuals for Oensingen(a) were associated with growth of newly seeded crops or cover crops following harvest. For example, in Dietiker et al. (2010), the second Oensingen crop 2005 (a Phacelia-based cover crop mixture) was modelled as a grassland with 50% above ground and 50% below ground biomass. These crops were not modelled by SPAc, and so corresponding residuals can be ignored in further discussion. The residuals in the pre-harvest phase do not show a clear pattern among sites. For instance, the model tended to estimate weaker sinks at Oensingen than the data in the month before harvest. In Auradé and Gebesee, the opposite was true, with the model suggesting a stronger sink than was measured pre-harvest. As expected, residuals were closest to 0 in the winter months when C fluxes generally are minimal, and (with exceptions) highest in the pre-harvest growing period.

3.1.2. Statistical description of daily NEE, and hourly NEE flux data

A linear fit between observed and modelled daily NEE (gC m^{-2} d^{-1}, sowing to harvest only) indicated that the R^2 was lowest for Oensingen(b) (0.74), and highest for Auradé and Oensingen(a) (0.88 each, Table 4). The slope of the linear fit was lower than 1 for all sites, indicating model bias. As negative intercept values suggested, sink strength was generally overestimated. Cumulative NEE (between sowing and harvest) were almost always lower than what had been observed, with a mean difference of ∼−134 gC m^{-2}. The R^2 of observed vs. modelled cumulative NEE for all site years was 0.27. RMSE was lowest for Auradé and highest for Oensingen(b), and ranged between 1.15 and 1.85 gC m^{-2} d^{-1}.

Five days (15–20.05.06) within growing period Auradé (ww) were chosen to display half-hourly values of modelled and simulated NEE (Fig. 6). The observed diurnal cycle of NEE, with the ecosystem's net C uptake during daytime and net C release C dur-

Table 3
List of model parameters, units, nominal values, and corresponding sources. For V_{cmax}, J_{max}, and C_{la}, values are given for winter wheat/winter barley. New parameters (SPAc vs. SPA) are marked (*), and all parameter values but I and t_{ar} have been altered.

Parameter symbol	Name	Unit	Nominal value	Source
N_{frac}	Leaf nitrogen distribution through canopy layers	Fraction	0.33, 0.27, 0.22, 0.18	Hirose and Werger (1987)
g_{plant}	Stem conductance	mmol m^{-2} s^{-1} MPa^{-1}	5	Adjusted to match leaf specific conductance from Liu et al. (2005)
Ψ_1	Minimum leaf water potential	MPa	−1.9	Johnson et al. (1987)
I	Stomatal efficiency	–	1.007	Adjusted to maintain max. g_s <400 mmol m^{-2} s^{-1} (Ye and Yu, 2008)
C	Leaf capacitance	mmol m^{-2} MPa^{-1}	2000	Estimated (Williams et al., 1996)
R_r	Root resistivity	MPa s g mmol^{-1}	10	Adjusted to match leaf specific conductance from Liu et al. (2005)
V_{cmax}	Maximum carboxylation capacity	μmol m^{-2} s^{-1}	64/79	Wullschleger (1993), Tambussi et al. (2005)
J_{max}	Maximum electron transport rate	μmol m^{-2} s^{-1}	137/157	Wullschleger (1993), Tambussi et al. (2005)
C_{la}	Carbon per leaf area	gC m^{-2}	19.5/15	Penning de Vries et al. (1989, p. 100)
r_{dc}	Decomposition rate	h^{-1}	2.3×10^{-5}	Buyanovsky and Wagner (1987)
f_a	Fraction of GPP respired	Fraction	0.44	Monje and Bugbee (1998)
t_{root}	Turnover rate of roots	h^{-1}	6.25×10^{-3}	Penning deVries et al. (1989, p. 210)
m_{lit}	Mineralization rate of litter	h^{-1}	2.8×10^{-4}	Buyanovsky and Wagner (1987)
m_{SOM}	Mineralisation rate of SOM/CWD	h^{-1}	2.28×10^{-6}	Buyanovsky and Wagner (1987)
t_{lab}	Turnover rate of labile pool	h^{-1}	6.25×10^{-3}	Penning de Vries et al. (1989, p. 47/48)
r_{tr}	Respiratory cost of labile transfers	Fraction	0.2133	Goudriaan and Van Laar (1994, p. 52)
t_{ar}	Turnover rate of autotrophic respiration pool	h^{-1}	0.07	Adjusted to give ∼daily turnover of pool
GDD_{em}*	Temperature sum at emergence	Degree days	125	Wang and Engel (1998)
trl_{stem}*	Rate of translocation of remobilisable carbon from stems	h^{-1}	8.3×10^{-3}	Penning de Vries et al. (1989, p. 47)
$r_{max,v}$*	Maximum development rate in vegetative phase	d^{-1}	0.04	Yan and Wallace (1998), Li et al. (2008)
$r_{max,r}$*	Maximum development rate in reproductive phase	d^{-1}	0.035	Streck et al. (2003)
T_{min}*	Minimum temperature for development		0	Li et al. (2008)
T_{opt}*	Optimum temperature for development		24	Li et al. (2008)
T_{max}*	Maximum temperature for development		35	Li et al. (2008)
$T_{min,vn}$*	Minimum temperature for vernalization		−1.3	Porter and Gawith (1999)
$T_{opt,vn}$*	Optimum temperature for vernalization		4.9	Porter and Gawith (1999)
$T_{max,vn}$*	Maximum temperature for vernalization		15.7	Porter and Gawith (1999)
VD_h*	Effective vernalization days when plants are 50% vernalized	VD	22.5	Streck et al. (2003)
LAI_{cr}*	Critical leaf area index beyond which leaf senescence due to self-shading occurs	m^2 m^{-2}	4	Van Laar et al. (1997)
dsh_{max}*	Maximum value of relative death rate due to shading	h^{-1}	1.25×10^{-3}	Van Laar et al. (1997)
PH_{cr}*	Minimum (or critical) photoperiod for development	h	8.25	Streck et al. (2003)
PH_{sc}*	Photoperiod sensitivity coefficient	–	0.25	Streck et al. (2003)

ing night-time, was reproduced by SPAc with high accuracy, both in terms of magnitude and timing. Moreover, elevated vapour pressure deficit (∼1.8 kPa in early afternoon hours on the 17th) and considerably reduced short-wave radiation levels (∼100 W m^{-2} around noon on the 18th) coincided with midday depressions in C assimilation rates, which were recognisable in plotted values of both observed and modelled NEE. The maximum temperature on the 17th was ∼10 higher than on the following day. Peak C assimi-

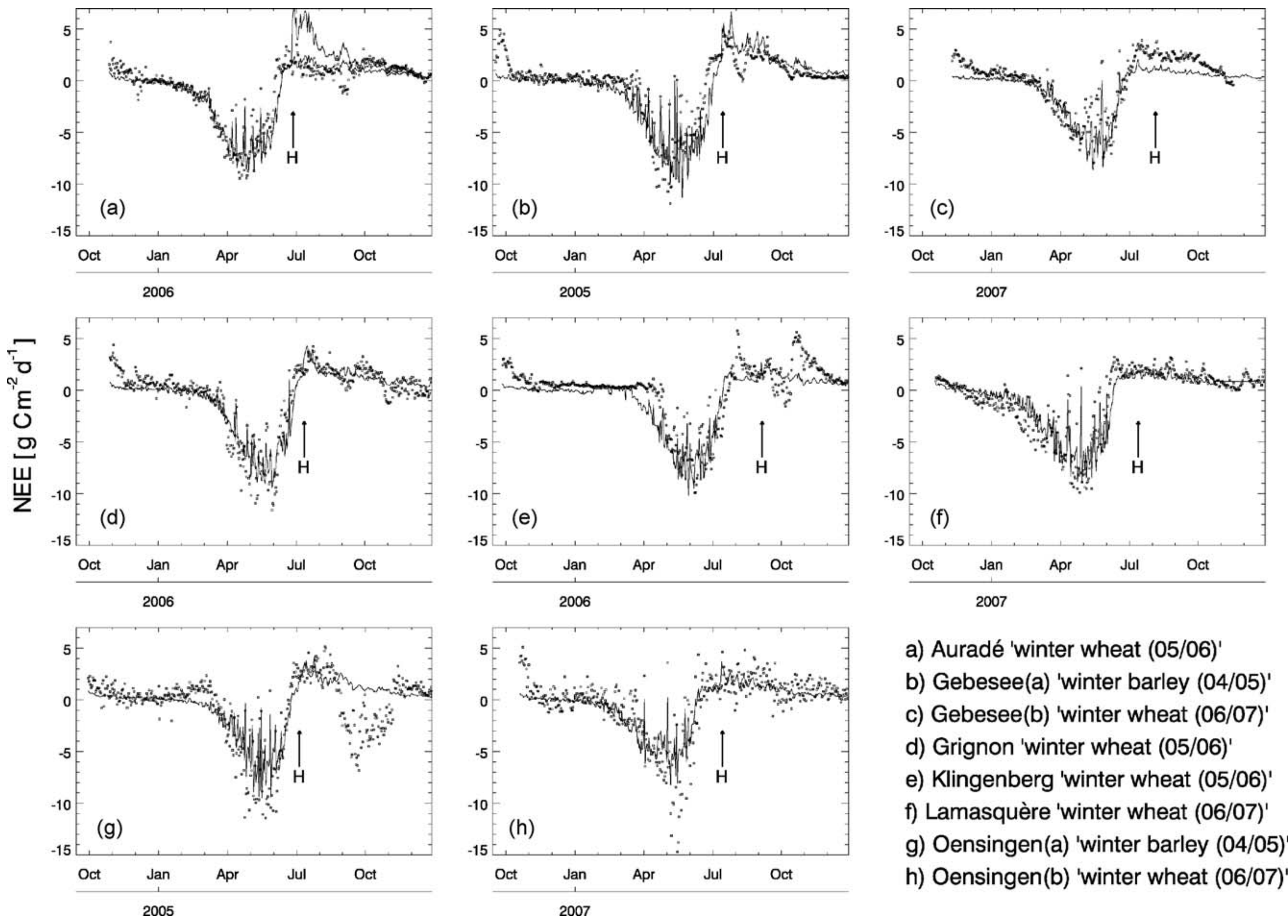

Fig. 4. (a–h) Observed (boxes) and modelled (solid line) daily values of NEE in $gC\,m^{-2}\,d^{-1}$ for each of the study sites and years. The timing of harvest (H) is indicated by arrows. Plotted values begin at sowing date as reported. In panel (a), two lines are shown for model results with 90% (lower NEE at harvest) and 0% (higher NEE at harvest) of the above-ground C mass (except for grains, which are always entirely exported) removed from the site at harvest.

lation rate seemed to be slightly overpredicted by SPAc for each of the 5 days.

3.2. Results for modelled biometric data

We compared simulations of leaf area index (LAI) and leaf C mass with independent measurements for all growing periods, except for Oensingen(a) and (b), for which no time series of observations of both LAI and leaf dry mass were available. Moreover, observed total leaf C mass values (i.e. not reported separately as both live and dead leaf C mass, except for Gebesee(a) + (b)) will contain a certain share of live and dead leaf C, depending on the crop developmental stage. Presumably, all reported values of total leaf C for the vegetative crop phase are live leaf C mass. During the reproductive phase with commencing and accelerating senescence, the fraction of dead leaf C contained in reported total leaf C values will increase continuously towards maturity, when finally all observations are for dead leaf C only. We have time series of observed total leaf C mass for Auradé, Grignon, Klingenberg, and Lamasquère, and both live and dead leaf C mass for Gebesee(a) + (b). We subsequently converted leaf dry mass to leaf C mass assuming the same fixed leaf C content of 0.459 (Penning de Vries et al., 1989) for total, live, and dead leaf dry mass. Various on-site measurements during the study periods indicated that leaf C content is relatively similar for both live and dead leaves. Similarly, we used C content fractions for stem, root, and grains (following Penning de Vries et al., 1989) to convert from reported values given as dry biomass per area to $gC\,m^{-2}$.

Table 4
Statistical description of modelling results (RMSE and linear fit between observed and simulated daily NEE from sowing to harvest, columns 2–5) and comparison of simulated vs. observed values of cumulative NEE (from sowing to harvest, $gC\,m^{-2}$), yield (gC), and LAI (columns 6–11).

Site	NEE comparison				$NEE_{harvest}$		Yield		LAI_{max}	
	RMSE	R^2	Slope	Intercept	Obs	Mod	Obs	Mod [% of obs]	Obs	Mod [% of obs]
Auradé	1.15	0.88	0.89	−0.82	−447	−551	283	239 [84]	3.13	7.05 [225]
Gebesee(a)	1.64	0.82	0.87	−1.05	−314	−603	292	240 [82]	3.36	6.43 [191]
Gebesee(b)	1.37	0.81	0.81	−0.81	−169	−359	387	205 [53]	3.24	5.66 [175]
Grignon	1.44	0.87	0.76	−0.80	−411.6	−526	350	230 [66]	6.20	6.56 [106]
Klingenberg	1.55	0.80	0.87	−0.90	−275.8	−482	318	257 [81]	3.22	6.49 [202]
Lamasquère	1.28	0.84	0.80	−0.35	−529	−524	394	253 [64]	4.48	6.89 [154]
Oensingen(a)	1.44	0.88	0.67	−0.55	−357	−397	287	184 [64]	4.30	5.05 [117]
Oensingen(b)	1.85	0.74	0.58	−0.68	−339	−387	255	213 [83]	5.40	6.08 [113]

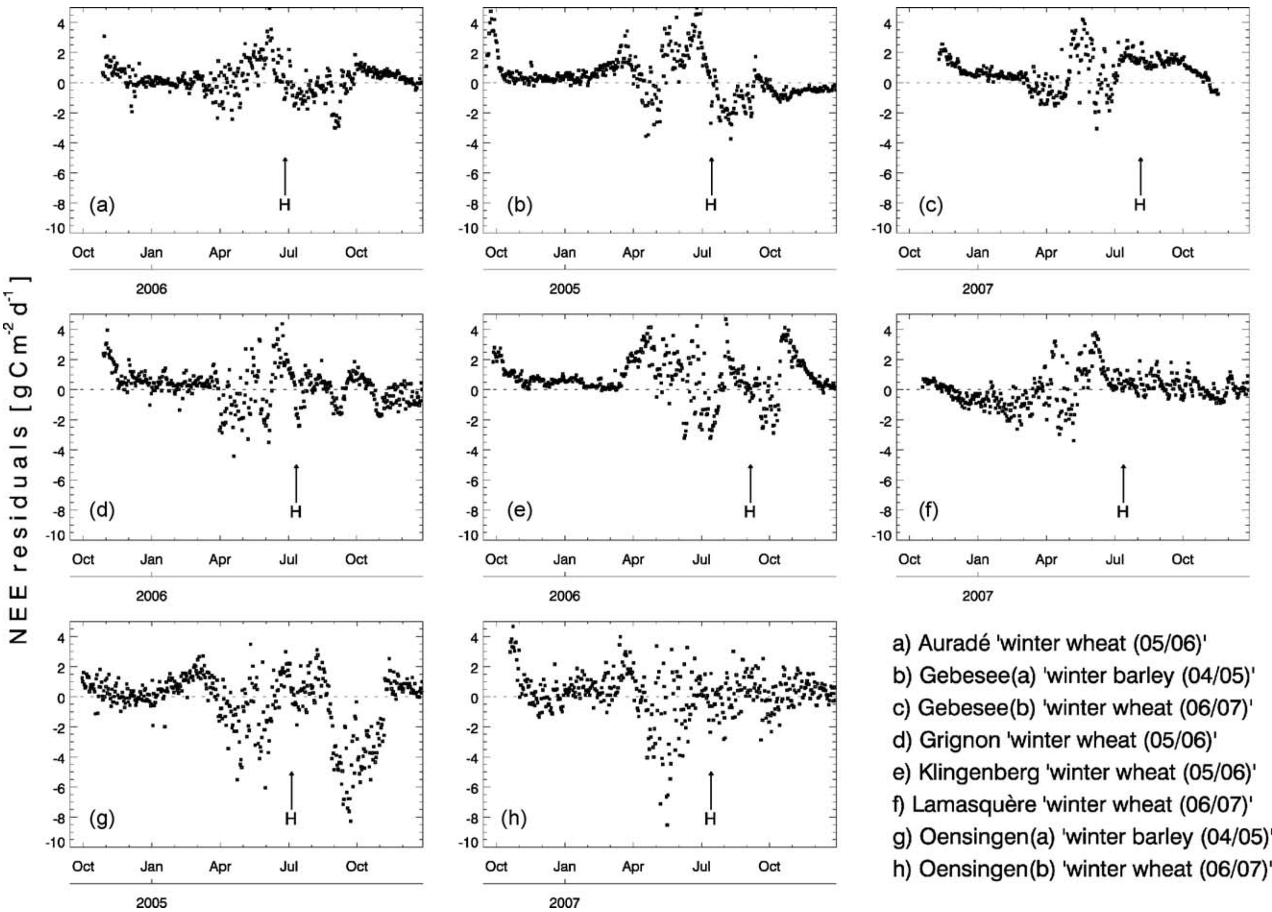

Fig. 5. (a–h) Observed minus modelled values of NEE (=NEE residuals) in gC m^{-2} d^{-1} for each of the study sites and years. The timing of harvest (H) is indicated by arrows.

SPAc modelled the dynamics of LAI and leaf C mass with high accuracy overall (mean R^2 values: 0.58 (LAI), 0.41 (live leaf C mass), 0.36 (total leaf C mass), Fig. 7). However, there was a greater agreement between predicted and observed peak leaf C mass than LAI values. There was a good match between observed and modelled LAI for the Grignon site, whereas modelled LAI was considerably higher than measured for all of the four remaining field sites. For Auradé, observed LAI values were ~50% lower than simulated. The timing of peak LAI and also senescence was represented well by the model for all but one of the sites: the modelled onset of the peak growth phase of LAI and leaf C was 20–30 days earlier than observed for Klingenberg.

Modelled live leaf C mass values were close to measurements of total leaf C during the vegetative phase (i.e. before the peak of LAI and leaf C), often within the uncertainty range of ±1 standard deviation (rows 2 and 4 in Fig. 7). They decreased considerably towards maturity as senescence accelerated, and just before harvest were 0 for all sites. In contrast to that, observed total leaf C mass values decreased at a slower rate and thus maintained higher values during the reproductive phase throughout. At harvest time, total leaf C mass typically was ~50% of its peak value (panels b, h, and l). Only for Klingenberg (panel j) the total leaf C mass dropped to a very low value (~20 gC m^{-2}) at crop maturity. The observed pattern of total leaf C mass decrease during the reproductive phase is bet-

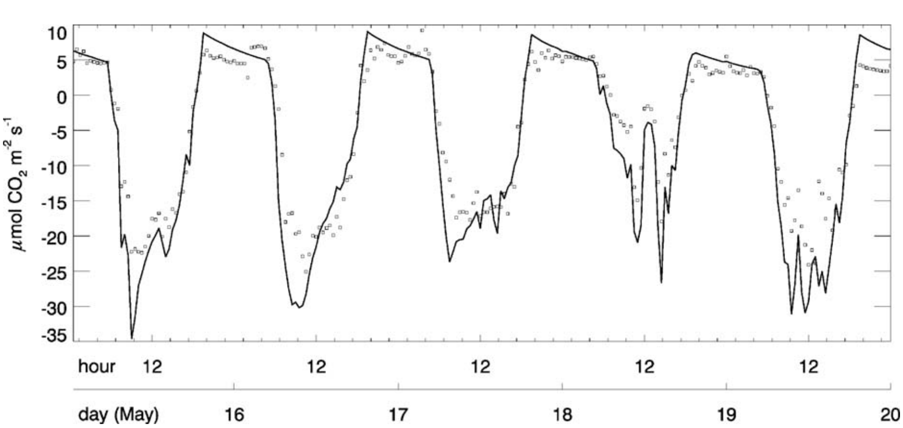

Fig. 6. Observed (boxes) and modelled (solid line) half-hourly values of NEE in μmol CO_2 m^{-2} s^{-1} for the 15–20 May 2006, Auradé (ww).

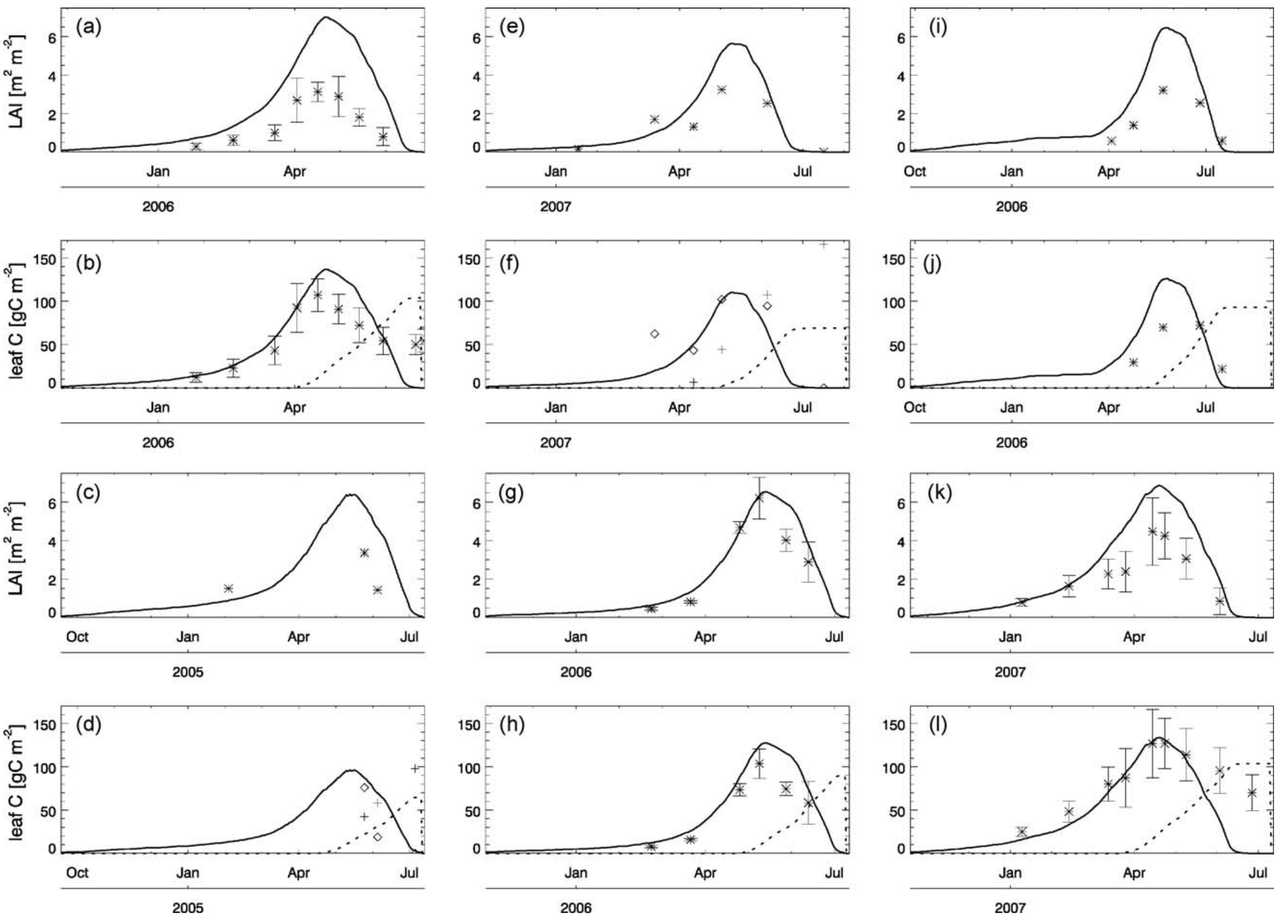

Fig. 7. (Rows 1 and 3) Modelled (solid line) and observed (*) values of LAI. (Rows 2 and 4) Modelled live (solid line) and standing dead (dashed line) leaf carbon mass values, together with observed time series of total (*), live (◊), and dead (+) leaf carbon mass. Error bars are ±1 standard deviation, where available. (a and b) Auradé, (c and d) Gebesee(a), (e and f) Gebesee(b), (g and h) Grignon, (i and j) Klingenberg, (k and l) Lamasquère.

ter matched when considering modelled standing dead leaf C mass (dashed line). Dead leaf C mass increased from the late vegetative phase until maturity, and, even though it then was considerably larger than observed values of total leaf C, was able to more closely follow the observed pattern. For Gebesee(a) + (b), observations of both live and dead leaf C mass are shown (panels d and f), indicating that dead leaf C mass accumulation starts earlier and reaches higher at-harvest values than modelled.

Observed values of peak LAI ranged from ~3.1 to ~6.2 m^2 m^{-2} compared to a modelled range from ~6.5 to ~7 m^2 m^{-2}. On average, modelled peak LAI was 60% higher than measured. Observed peak leaf C mass data ranged from ~70 to ~125 gC m^{-2} compared to simulated values ranging from ~125 to ~140 gC m^{-2}. On average, modelled peak live (total) leaf C mass was 24% (44%) higher than measured.

Yield was always underestimated by SPAc for all field sites and years (harvested grain C mass in gC as reported from the site, columns 8 and 9 in Table 4: 71% of observed, on average), whereas maximum LAI was continuously higher than the measurements in field (columns 10 and 11: 160% of observed, on average). The R^2 of observed vs. modelled yield for all site years was 0.05, with no significant correlation.

3.3. Sensitivity analysis

The model showed highest sensitivity to the parameters that determined the fraction of GPP allocated to autotrophic respiration (f_a) and the minimum photoperiod for development (PH_{cr}) (Table 5). The nominal value of cumulative NEE at harvest was −550.62 gC m^{-2}. Considerable model sensitivity was evident for parameters describing the leaf mass per area (C_{la}), the maximum development rate in both the vegetative and reproductive phases ($r_{max,vr}$), the parameters relating temperature to development ($T_{min/opt/max}$), and the photoperiod sensitivity coefficient (PH_{sc}). Most of these parameters were applied within the crop developmental module. Model sensitivity was particularly low to vernalization-related parameters and to those parameters used for estimating the rate of decomposition and mineralization processes.

Model sensitivity to the distribution of leaf nitrogen (N_{frac}) through the four canopy layers (not listed in Table 5) was tested by comparing the leaf N distribution based on Hirose and Werger (1987) with a uniform distribution, resulting in a new cumulative NEE of −523.17 gC m^{-2} (a relative change of 5%).

4. Discussion

4.1. Modelled carbon exchange data

For all of the study years presented, SPAc was able to simulate the seasonality of C fluxes with high accuracy (mean R^2 = 0.83, daily NEE data between sowing and harvest of all growing periods). This seasonality greatly depends on the developmental model and C allocation and remobilization patterns associated with it (Table 5). Seeds germinated soon after sowing in autumn, so SPAc simulated (rather low) photosynthesis and gradual growth of LAI during

Table 5
Results of sensitivity analysis for each of the parameters in Table 3 except for N_{frac} (in text). Nominal NEE value (NEE_{nom}) is −550.62. Columns 2 and 3 contain values of absolute change in NEE (i.e. $\Delta NEE = NEE - NEE_{nom}$), both for a 25% increase and decrease of the parameter concerned. Columns 4 and 5 contain values for sensitivity coefficient *S*. For the cardinal temperatures of development, each parameter was altered by ±2 and no *S* value could be calculated.

Parameter	ΔNEE (%)		*S*	
	+25%	−25%	+25%	−25%
g_{plant}	−4.04 (−0.7)	6.63 (1.2)	0.03	00.5
Ψ_1	−44.1 (−8.1)	70.20 (12.7)	0.32	0.51
I	4.79 (0.9)	−4.34 (−0.8)	−0.03	−0.03
C	−5.57 (−1.0)	5.66 (1.0)	0.04	0.04
R_r	42.07 (7.6)	−44.46 (−8.1)	−0.31	−0.32
V_{cmax}	−10.95 (−2.0)	28.11 (5.1)	0.08	0.20
J_{max}	−94.04 (−17.1)	111.65 (20.3)	0.68	0.81
C_{la}	111.65 (20.3)	−158.92 (−28.9)	−0.81	−1.15
r_{dc}	−0.08 (0.0)	0.08 (0.0)	0.00	0.00
f_a	264.97 (48.1)	−261.33 (−47.5)	−1.92	−1.90
t_{root}	2.72 (0.5)	−2.76 (−0.5)	−0.02	−0.02
m_{lit}	2.02 (0.4)	−2.34 (−0.4)	−0.01	−0.02
m_{SOM}	25.02 (4.5)	−25.45 (−4.6)	−0.18	−0.18
t_{lab}	−20.30 (−3.7)	31.20 (5.7)	0.15	0.23
r_{tr}	17.76 (3.2)	−18.06 (−3.3)	−0.13	−0.13
t_{ar}	0.00 (0.0)	−0.01 (0.0)	0.00	0.00
GDD_{em}	0.00 (0.0)	0.00 (0.0)	0.00	0.00
trl_{stem}	2.03 (0.4)	−2.15 (−0.4)	−0.01	−0.02
$r_{max,v}$	76.51 (13.9)	−103.85 (−18.9)	−0.56	−0.75
$r_{max,r}$	33.59 (6.1)	−49.00 (−8.9)	−0.24	−0.36
T_{min}	−35.62 (−6.5)	28.39 (5.2)		
T_{opt}	−150.81 (−27.4)	152.11 (27.6)		
T_{max}	63.01 (11.4)	−75.91 (−13.8)		
$T_{min,vn}$	−0.62 (−0.1)	0.22 (0.0)		
$T_{opt,vn}$	0.24 (0.0)	−0.61 (−0.1)		
$T_{max,vn}$	0.66 (0.1)	−0.74 (−0.1)		
VD_h	−1.84 (−0.3)	3.16 (0.6)	0.01	0.02
LAI_{cr}	−15.59 (−2.8)	26.99 (4.9)	0.11	0.20
dsh_{max}	6.59 (1.2)	−7.74 (−1.4)	−0.05	−0.06
PH_{cr}	−159.27 (−28.9)	101.14 (18.4)	1.16	0.73
PH_{sc}	40.60 (7.4)	−60.30 (−11.0)	−0.29	−0.44

autumn and winter. In the winter and early spring months, photosynthesis was limited by low temperatures and global irradiation levels, which SPAc was able to represent. However, throughout late spring and summer, the crop plants matured relatively quickly. As a consequence, only a short peak C assimilation phase was observed, followed by rapid senescence, which the modelling captured effectively (Fig. 4).

We define winter cereal "dormancy" as a period of relatively low rates of photosynthetic activity during autumn and winter, rather than a period of non-photosynthetic-activity. Observed values of hourly NEE data show a daily pattern of C assimilation and release during the entire growth period, including autumn and winter months (data not shown). Accordingly, we did not add a temperature-dependant "dormancy" parameter in order to additionally limit photosynthesis in SPAc during winter.

As the ecosystem's assimilatory capacity is closely related to its developmental stage through shifting C allocation patterns, remobilization, and senescence, the modelling approach adopted within SPAc is able to represent typical winter wheat/barley C flux dynamics. Growing period Klingenberg was exceptional, as the observed onset of the peak growing phase occurred about 1.5 months later than modelled. Moreover, due to a rainy period from end of July to end of August 2006, this season's harvest date was 1–2 months later than those of the other sites. Currently, SPAc is not accounting for a delaying factor (probably in order to limit photosynthesis rather than development) that could improve model results for this single growing season.

SPAc was closest to observations during the winter and early spring months (the "dormancy" phase), whereas daily NEE was lower than observed (i.e. C assimilation overestimated) for most of the sites during the peak growth phase itself. Correspondingly, for all but one of the growing seasons (Lamasquère (ww) was this exception) the modelled cumulative NEE at harvest was lower than observed, and all intercept values of lines fitted to observed vs. modelled scatter plots were negative (Table 4). The R^2 value of 0.27 of observed vs. modelled cumulative NEE (sowing to harvest) for all site years further indicates that SPAc needs to be improved in order to allow for an enhanced representation of cumulative NEE for different years and site locations. We expect major improvements by increasing the initial litter C content (thus increasing post-sowing heterotrophic respiration), as observed NEE was generally higher than modelled in the days following sowing. The seasonality of C exchange was mainly a function of the crop's maturity (developmental stage), but the magnitude of daily NEE was largely determined by the photosynthesis-related modules within SPAc. Correspondingly, adjusting related parameters, for instance those describing carboxylation rate (V_{cmax} and J_{max}), will likely improve model performance. This overprediction of net C assimilation is further confirmed through the modelled hourly NEE data for Auradé (Fig. 6), which in daytime was almost always more negative than observed. However, SPAc simulated the effects of high vapour pressure deficit (17.05.06) and low global irradiation levels (18.05.06) effectively.

SPAc predictions of NEE are only truly representative of the period between sowing and harvest, as no post-harvest regrowth of volunteers or land management actions have been considered. This largely explains differences between modelled and observed post-harvest NEE values (Fig. 5). For example, manure application in October 2006 (~176 gC m^{-2}) could explain high observed values of NEE for Klingenberg, and the sowing of a cover crop (*Phacelia* sp., *Avena* sp., and *Trifolium alexandrinum*; 20.07.05) caused a later second C assimilation phase observed at Oensingen(a).

The reported fraction of crop biomass remaining on field after harvest was considered in all SPAc model runs to simulate realistic amounts of crop residue decomposition and mineralisation (i.e. heterotrophic respiration). During the growing period at Auradé, only the grains have been exported, and the remaining aboveground biomass was left on the field after harvest. Due to this large litter flux, SPAc initially modelled a considerable contribution of decomposing crop residue to post-harvest respiration, further increasing NEE for this period (thin solid line in panel (a) of Fig. 4). However, NEE observations did not show any large amount of crop residue decomposition and mineralisation after harvest. The crop residue was incorporated into the soil at ploughing on the 30.09.06. As summer precipitation had been relatively low, the residues might have dried out on the surface, and consequently, decomposition might have not started before ploughing. Additionally, small voluntary regrowth occurred after harvest, which would have further influenced post-harvest NEE (E. Ceschia, personal communication). Thus, model results were much closer to observations with a parameterisation where only 10% of the above-ground biomass (except for grains) remained on the site after harvest. More data are needed in order to improve the limited understanding of how land management influences crop residue decomposition and mineralisation rates. In the crop growth model APSIM, decomposition rate is set to decrease as the crop residue dries up, based on potential evaporation (Keating et al., 2003). Data on direct observations of soil respiration fluxes would further help to constrain the estimation of heterotrophic respiration during and outside of crop growth periods, as proposed by Wang et al. (2005).

4.2. Modelled biometric data

Except for Grignon, SPAc generally overestimated LAI when compared to observations (R^2 of observed vs. modelled LAI_{max} was <0.01), but simulated values of leaf C compare considerably better to measurements (R^2 of observed vs. modelled maximum leaf C was 0.33, Fig. 7). As not only LAI, but also (though to a smaller extent) live leaf C mass are often overpredicted by SPAc, the fraction and/or duration of C allocation to leaves might be too high. Alternatively, increasing parameter C_{la} (C per leaf area) would make it more "C expensive" to grow leaves, thus reducing LAI. Due to high model sensitivity, a change in C_{la} would certainly make an impact on overall C assimilation capacity. However, very similar total leaf C mass values have been observed at Auradé and Grignon, but there are large differences in LAI. This difference could be explained by either alternative methods of measuring LAI (due to the contributions of green stems and ears, the LAI measured above the canopy can be quite different to the LAI measured through destructive green leaf sampling; Hoyaux et al., 2008) or different C per leaf area (C_{la}) values. Another potential error source in this model is the assumption of a constant C_{la} over the growing period (Kucharik and Twine, 2007).

In other studies, observed values of LAI_{max} for winter wheat were ~3 (Lokupitiya et al., 2009), 4–6 (Wang et al., 2007), and ~5 m m^{-2} (Arora, 2003; Wang et al., 2005). In Huang et al. (2009), measured LAI_{max} correlated positively with synthetic fertilizer N application rates (0–200 kg N ha^{-1}), and ranged from 1 to 6.5 m m^{-2}. Their data suggest that we might be predicting LAI for winter cereals under optimal growth conditions. Thus, a more detailed consideration of cultivar-specific parameters affecting LAI, but also land management issues such as fertilization, pest/weed control, and irrigation would result in a more realistic prediction of LAI for various sites and years. Another improvement to SPAc would be to account for the contribution of green stems and ears to overall photosynthesis. In SUCROS, a mechanistic crop growth model on which the modelling approach adopted in SPAc is greatly based upon, ear photosynthesis is accounted for by assuming half of the ear area index (EAI, in m m^{-2}) to contribute to overall light absorption. Moreover, the significance of not only green ear, but also stem photosynthesis, has been demonstrated in Hoyaux et al. (2008). Their study, a model-data comparison of C flux data for winter wheat in Lonzée, Belgium, showed that highest model uncertainty resulted from varying C assimilation rates of stems and ears. When only considering green leaf photosynthesis, cumulative GPP was underestimated by 23%. If stem and ear photosynthesis were considered in SPAc, leaf C mass, and as a consequence LAI, could be reduced without considerably reducing the crop plant's assimilatory capacity.

For Klingenberg, we modelled an earlier onset of peak LAI and leaf C mass growth than observed. As this lag can also be seen in the daily flux data, SPAc seems to have overpredicted photosynthesis in the early spring months, leading to an increase in LAI and leaf C mass not supported by measurements made on site. SPAc has also slightly overpredicted GPP during the winter months, resulting in a (rather gradual) build-up of foliar biomass during this period. By the onset of the peak growth phase in spring, this winter growth could have resulted in too high initial values of LAI and leaf C mass, allowing for higher than observed photosynthetic rates. In this context, measurements of LAI or leaf DM made during the "dormancy" phase would help to further explain and constrain model behaviour.

Compared to SPAc runs with one single foliar C pool, the addition of a standing dead foliage C pool led to an improved prediction of at-harvest total foliar C mass, which was measured to be considerably >0 for most of the sites (Fig. 7). At maturity, the observeable leaf C of winter cereals was predominately standing dead leaf C mass, which remained after leaf mass loss through senescence and remobilization. Without a modelled standing dead foliage C pool, leaf C mass at maturity would be largely underestimated by SPAc. Final dead leaf C mass is higher than observed values of total leaf C at harvest for all sites, but were considerably lower than measured dead leaf C at Gebesee(a) + (b).

Leaf senescence is triggered in SPAc at an LAI of ≥4, thus the dead foliage C pool began to receive C whilst the alive foliar C mass was still growing. This is realistic, as self-shading can induce senescence of lower leaves while the foliar biomass in upper levels might still be increasing. An alternative way of modelling leaf senescence would be to connect LAI to leaf N content: leaves will begin to senesce when the leaf N concentration drops below a value required for photosynthesis (Kucharik and Twine, 2007).

Based on the observed data available, it is difficult to conclude whether we are actually under- or overestimating dead leaf C mass at maturity. Whichever is true, the model could be improved by changing the conditions of triggering leaf senescence or by altering the fraction of senescing leaf biomass allocated to the dead foliage C pool (currently 0.5, the other half being treated as remobilized C, which will finally be allocated to the storage organ).

In another study, leaf senescence was found to be poorly represented, resulting in a considerable mismatch between observed and simulated LAI (Kucharik and Twine, 2007). Improving the representation of crop senescence in agroecosystem modelling is of further importance: standing dead C mass during the reproductive phase, but more importantly the post-harvest crop residue layer, together have the potential to improve the surface energy balance when considered in simulations, as they affect surface albedo and other physical properties (Kucharik and Twine, 2007). Moreover, an advanced representation of crop senescence and its effect on vegetation greenness will increase the usability of remote sensing data (e.g. MODIS 250m NDVI) for data assimilation and validation. We see in the modelling approach followed here a step forward in representing leaf senescence and estimating standing dead leaf C mass for these purposes.

We might expect that with a stronger simulated C sink than measured, the simulated grain yield would be an overestimate. However, model simulations actually underestimated grain yield in all cases by a mean value of ~100.4 gC m^{-2} (72% of observed), and the R^2 value for all site years was 0.05. The biometry and flux residuals seem to suggest that less C is allocated to foliage. If this were so, then leaf C stocks would be reduced, and C sink strength would be decreased as a result of resultant reductions in GPP. However, of the smaller amount of C assimilated, a larger fraction and larger absolute amount must be allocated to grain filling. Data on root allocation and turnover would be valuable to better constrain below ground allocation and resolve these discrepancies. Moreover, the already discussed consideration of green ear and stem photosynthesis would allow for lower leaf C allocation without considerably reducing total C assimilation.

There are different approaches in the crop growth modelling community in predicting yield. For example, growth of the storage organs is source-limited in the EPIC model, whereas it is both source and sink-limited in CERES (Mearns et al., 1999). However, the questions remain whether grain number is the controller of yield formation (sink limited) or rather a reflection of yield formation (source limited) (Jamieson et al., 1998): the Sirius crop growth model successfully predicted yield without an explicit dependence on grain population. In SPAc, yield growth is both source and sink-limited.

Improving the simulation of cumulative NEE, LAI, and yield will be the focus of future research with SPAc, as these entities are currently represented with relatively low accuracy by the model.

4.3. Which parameters are most sensitive?

We found high model sensitivity to a range of different parameters, which can be distinguished into two main groups: photosynthesis-related and development-related parameters. In the first group, parameter f_a (the fraction of GPP respired) is a major control on model sensitivity, as it largely determines the amount of C available for growth. Indeed, it is known that f_a not only varies over a relatively broad range for crops (0.3–0.6 for maize, rice, and wheat), but also that it changes with crop development, rather than being constant throughout the plant's life span (Amthor, 2000). Crop f_a is lower than for natural vegetation, as the crop breeders' selection of genotypes with large harvest indexes may indirectly have led to the selection of reduced autotrophic respiration (Amthor, 2000). We conclude that both model sensitivity and parameter uncertainty are high for f_a. Our simulations of autotrophic respiration fluxes might be improved if we more explicitly simulated growth and maintenance respiration as separate processes, an approach with a long tradition in models of higher-plant respiration (Amthor, 2000). However, this change would come at the cost of increasing model complexity set against a lack of detailed data constraints. Moreover, formulations of growth and maintenance respiration are purely notional functional constructs, and they are not biochemically distinct from each other. Their coefficients are not fundamental but are defined only operationally by a particular measurement protocol and assumptions associated with it (Gifford, 2003). Thus, the philosophy of modelling autotrophic respiration in SPAc was chosen to be more simplistic.

The amount of leaf C per area (C_{la}) is another important parameter with a high influence on crop canopy development, and correspondingly high model sensitivity. Maximum electron transport rate (J_{max}) is directly used within SPAc's photosynthesis module and thus a crucial factor in estimating the amount of C available for growth. Additional studies would help in determining to which extent these parameters are cultivar dependent and thus might explain site-to-site and year-to-year differences in the magnitude of overall C assimilation through one crop type. Comparably low model sensitivity was evident for parameters used for modelling plant hydraulics, reflecting a lack of water stress impacts on C dynamics and plant development. SPAc was also relatively insensitive to changing decomposition and mineralisation rates, although the importance of residue management was clearly demonstrated. As the model's insensitivity to parameter GDD_{em} suggests, it seems to be unlikely that SPAc is considerably sensitive to potential shortcomings in the model structure itself. These shortcomings are still present as a lack of detail in simulating seed germination and growth of roots and shoots until emergence.

We also found high model sensitivity to a range of development-related parameters. In particular those coefficients estimating the influence of day length (photoperiod) on developmental rate play an important role in establishing developmental stage and thus shifting C allocation patterns, senescence and C remobilization. Even though some modern crop varieties are known to have little or no sensitivity to day length (to facilitate management over various cropping systems and larger regions), this sensitivity is still a desirable trait to keep harvest dates constant in areas of largely varying sowing dates (Penning de Vries et al., 1989). Photoperiod sensitivity is still the norm and a very powerful adaptive mechanism (Craufurd and Wheeler, 2009). Together with maximum developmental rates in both the vegetative and reproductive phases, photoperiod-related parameters exhibit notable model sensitivity in SPAc and thus need to be well constrained. We found considerable model sensitivity for non-vernalisation related cardinal temperatures of development, highlighting that a well defined set of parameters is of paramount importance. An improvement could be to use different cardinal temperatures for various sub-phases, e.g. as shown in Porter and Semenov (2005) and Streck et al. (2003). Temperature is crucial in determining the duration of important phases such as leaf growth (C assimilation/vegetative phase) and grain filling (senescence/reproductive phase). There already exists a set of studies that provide estimates of these cardinal temperatures (Li et al., 2008; Porter and Semenov, 2005; Streck et al., 2003; Wang and Engel, 1998; Xue et al., 2004), however these still vary considerably in relation to the number of sub-phases and cardinal temperature values themselves. Especially in the context of regional studies, which require generic estimates of these parameters for one single crop type, it is important to have a better estimate of the full range of parameter space. In SPAc, a 2 cardinal temperature change results in a considerable change in NEE (Table 5). More research is needed in order to determine whether the application of various different sets of cardinal temperatures (one for each individual sub-phase as in Porter and Semenov (2005) and Streck et al. (2003)) is superior over one/two single parameter sets which are constant from emergence to maturity. An increased number of cardinal temperature sets would result in increased model accuracy but also complexity. The temporal and spatial scale at which a certain model is applied will largely determine what is more important: accuracy or simplicity.

4.4. Is the modelling of developmental stages reliable?

The simulation of crop developmental stages is a novel part of SPAc and a key element in controlling fundamental model processes such as C partitioning, senescence, and remobilization. In another study, grain yield prediction was found to be much more dependent on the simulation of crop development and LAI than of the various components of yield, such as number of ears or grains (Jamieson et al., 1998). Unfortunately, no on-field observations of developmental or phenological stages have been made during the study periods presented here. We are currently developing and testing a methodology that would enable us to observe the timing of key crop developmental stages and land management operations from space, based on MODIS 250m NDVI data time series. These timings will probably be sowing date (summer crops), break of winter "dormancy" (i.e. the onset of the peak growth phase of winter cereals), and maximum LAI/flowering date. This new source of data would enable us to compare our estimates of crop development solely based on local meteorology with an independent means of data, and also provide input data for future regional modelling. The potential usefulness of MODIS data and their applicability for crop phenology detection have been highlighted in several studies (Huang et al., 2009; Kucharik and Twine, 2007; Sakamoto et al., 2005).

Overall, our results suggest that SPAc captures the typical seasonality of winter wheat/barley NEE reasonably well (Fig. 4), leading to the conclusion that the developmental model predicts the key timings of vegetation growth in a realistic manner. Growth during the winter period is restricted mainly through low temperatures and global irradiation, and thus photosynthesis. In all our model runs we see a small initial growth of leaves and roots soon after the sowing date, having accounted for a delay due to germination. At many sites, an initial high efflux of C has been observed around sowing, which then drops sharply. This C emission can possibly be explained by crop residue decomposition and/or ploughing shortly before sowing, which we are not considering in our model runs. The rapid decline of NEE can be seen as a result of initial growth of the winter crop sown.

In general, C assimilation during the vegetative phase appears to be controlled by photosynthesis as a function of local meteorology, whereas crop development causes the rapid increase of NEE in the reproductive phase. Crop development controls this shift from increasing to decreasing C assimilation through (1) a shift in C partitioning (from leaves and roots to stems and grains), and (2) the onset and acceleration of senescence as the crop matures. As all model runs reproduce the timing of these various stages with high accuracy (Fig. 4), we are confident that SPAc realistically models crop developmental stages.

Even though we are not accounting for any land management operations other than sowing and harvest, such as ploughing, fertilization, and application of herbicides, modelled NEE are still considerably close to observations. It is possible that some pests are responsible for the reduced productivity observed.

4.5. Is a generic winter wheat/barley parameterization acceptable?

SPAc effectively modelled the seasonality of NEE for winter wheat/barley over a range of western-central European climatic conditions and crop cultivars. Moreover, model inconsistencies were largely consistent across the different sites (a general overestimation of the C sink, Table 4), suggesting that small changes to the model parameterisation would improve results for all sites. It is debatable whether a few parameters should be isolated from this set in order to be allowed to vary with latitude/longitude or local climatic conditions. More research is needed for establishing the variability of cardinal temperatures, and especially photoperiod coefficients, for various spatial scales and climates.

A more cultivar-specific parameterization would certainly improve model results. However, it is desirable to keep model complexity low for larger scale studies, so a trade-off between model accuracy and model precision is unavoidable. This study demonstrated that a generic cereal parameterization allowed for rather accurate than precise predictions of cropland C fluxes over six different locations in western-central Europe. We are convinced that an improved *generic* parameterization, together with changes in model structure regarding green stem/ear photosynthesis, would result in considerably more precise simulations of both cereal C fluxes and stocks for western-central Europe. Overall, there was no evidence suggesting different parameters were required for wheat and barley.

4.6. Need for further measurements

The crop modelling approach followed here is largely based on, and compared with, a time series of biometric measurements. In this context, more data are required for model calibration and validation, especially regarding below ground partitioning coefficients and rates and determinants of senescence. Ideally, these empirical studies would be accompanied by observing the timing of key crop developmental stages. Also, as crop residue decomposition/mineralisation rates were found to be sensitive to land management, more C flux observations for varying amounts of crop residue biomass and land management operations would help to improve the representation of model processes in the post-harvest phase. Further parameters with high model sensitivity, but also high uncertainty (and thus requiring further research), are the fraction of GPP respired (f_a), the amount of leaf C per area (C_{la}), Farquhar parameters, developmental coefficients, and C allocation patterns.

5. Conclusions

We have demonstrated the coupling of a crop developmental model with a model of ecosystem C balance. The coupled model is capable of effectively describing the timing and magnitude of C exchanges for cereal crops in western-central Europe, compared against flux and biometric data. We found relatively high model accuracy for predicting daily C NEE fluxes over six different cropland sites and eight study periods (mean $R^2 = 0.83$, mean RMSE = 1.47 gC m^{-2} d^{-1}). However, SPAc simulated cumulative NEE (sowing–harvest) with lower accuracy (mean $R^2 = 0.27$). We conclude that the high quality of modelled daily NEE data is a result of realistically modelling cereal crop development, senescence, and remobilization. The simulation of crop development is a key new part of the model, making a crucial contribution in more realistically simulating C assimilation during the reproductive phase through shifts in C partitioning coefficients and the onset and acceleration of senescence. The general overprediction of photosynthesis is possibly a result of model weaknesses in terms of model parameterization and structure, leading to a larger C sink size than observed (mean cumulative NEE difference (observed − modelled) ∼ −133.9 gC m^{-2}). However, further data on fine root or below ground C allocation patterns is required in order to explain the overpredicted sink strength whilst underpredicting yield. As model error was found to be largely consistent over the various study sites, we conclude that an improved single parameterization will be able to produce realistic predictions of winter cereal C balance for western-central Europe.

Moreover, observed daily NEE data suggested active photosynthesis during the entire crop growth period, including the so-called winter "dormancy" months. SPAc was able to match these observations solely based on local meteorology and without further restrictions on photosynthesis, such as a threshold temperature value defining the duration of this "dormancy" period. This suggests that the concept of "dormancy", as it is known from natural ecosystems, is not directly applicable to winter cereal crops.

The model was also able to generally simulate realistic patterns of post-harvest decomposition and mineralisation fluxes as a function of crop residue C mass. However, these fluxes were clearly overestimated when no crop residue was exported from the field. We expect improved simulation of decomposition and mineralisation rates when accounting for the effects of crop residue moisture on these rates.

LAI was overestimated by 60% on average, indicating that we predict LAI for optimal non-stressed conditions, but yield was underestimated (∼71% of observed). We require a parameter calibration or model structure that will (1) decrease LAI, (2) slightly reduce live leaf C mass, and (3) increase yield whilst (4) decreasing the overall C assimilation strength. These changes could result from alterations to the parameter for C per leaf area and the dynamics of root allocation. Additionally, a significant improvement could be made to the model structure by accounting for the contributions of green ears and stems on overall photosynthesis, allowing for lowering LAI and live leaf C without lowering C assimilation more than necessary.

Another crop C budget model novelty was the inclusion of a dead leaf C pool, which allowed for a more realistic prediction of at-harvest above-ground C stocks. Moreover, we expect an improved simulation of the surface energy balance (mainly through albedo effects), and of optical properties of the crop canopy, which is important for future studies based on remote sensing data (e.g. MODIS 250m NDVI time series).

Further studies should focus on improving the development parameterisations for photoperiod and temperature, and ascertaining the allocation of photosynthate to autotrophic respiration. Additionally, more estimates of C per leaf area values but also C allocation patterns to both above and below ground organs of cereal crops are required.

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References

Adiku, S.G.K., Reichstein, M., Lohila, A., Dinh, N.Q., 2006. PIXGRO: a model for simulating the ecosystem CO_2 exchange and growth of spring barley. Ecological Modelling 190 (3–4), 260–276.

Amthor, J.S., 2000. The McCree-de Wit-Penning de Vries-Thornley Respiration Paradigms: 30 years later. Annals of Botany 86 (1), 1–20.

Anthoni, P.M., Freibauer, A., Kolle, O., Schulze, E.-D., 2004. Winter wheat carbon exchange in Thuringia, Germany. Agricultural and Forest Meteorology 121 (1–2), 55–67.

Arora, V.K., 2003. Simulating energy and carbon fluxes over winter wheat using coupled land surface and terrestrial ecosystem models. Agricultural and Forest Meteorology 118 (1–2), 21–47.

Aubinet, M., Moureaux, C., Bodson, B., Dufranne, D., Heinesch, B., Suleau, M., Vancutsem, F., Vilret, A., 2009. Carbon sequestration by a crop over a 4-year sugar beet/winter wheat/seed potato/winter wheat rotation cycle. Agricultural and Forest Meteorology 149 (3–4), 407–418.

Béziat, P., Ceschia, E., Dedieu, G., in press. Carbon balance of a three crop succession over two cropland sites in South West France. Agricultural and Forest Meteorology.

Bondeau, A., Smith, P.C., Zaehle, S., Schaphoff, S., Lucht, W., Cramer, W., Gerten, D., Lotze-Campen, H., Muller, C., Reichstein, M., Smith, B., 2007. Modelling the role of agriculture for the 20th century global terrestrial carbon balance. Global Change Biology 13 (3), 679–706.

Buyanovsky, G.A., Wagner, G.H., 1987. Carbon transfer in a winter wheat (*Triticum aestivum*) ecosystem. Biology and Fertility of Soils 5 (1), 76–82.

Chouard, P., 1960. Vernalization and its relations to dormancy. Annual Review of Plant Physiology 11 (1), 191–238.

Craufurd, P.Q., Wheeler, T.R., 2009. Climate change and the flowering time of annual crops. Journal of Experimental Botany 60 (9), 2529–2539.

de Noblet-Ducoudré, N., Gervois, S., Ciais, P., Viovy, N., Brisson, N., Seguin, B., Perrier, A., 2004. Coupling the soil-vegetation-atmosphere-transfer scheme ORCHIDEE to the agronomy model STICS to study the influence of croplands on the European carbon and water budgets. Agronomie 24, 397–407.

Denman, K.L., Brasseur, G., Chidthaisong, A., Clais, P., Cox, P.M., Dickinson, R., Hauglustaine, D., Heinze, C., Holland, E., Jacob, D., Lohmann, U., Ramachandran, S., da Silva Dias, P., Wofsy, S., Zhang, X., 2007. Climate Change 2007: the physical science basis. In: Couplings Between Changes in the Climate System and Biogeochemistry. Contribution of Working Group I to the Fourth Assessment Report of the Intergovernmental Panel on Climate Change. Cambridge University Press, Cambridge, New York.

Dietiker, D., Buchmann, N., Eugster, W., 2010. Comparing measured CO_2 and H_2O fluxes under real conditions and yield under changing climatic conditions on cropland with the DNDC model. Agriculture, Ecosystems & Environment 139, 396–401.

Eurostat, 2008. Agricultural Statistics. Main Results – 2006–2007. Eurostat Pocketbooks, Luxembourg.

Falge, E., Baldocchi, D., Olson, R., Anthoni, P., Aubinet, M., Bernhofer, C., Burba, G., Ceulemans, R., Clement, R., Dolman, H., Granier, A., Gross, P., Grünwald, T., Hollinger, D., Jensen, N.-O., Katul, G., Keronen, P., Kowalski, A., Lai, C.T., Law, B.E., Meyers, T., Moncrieff, J., Moors, E., Munger, J.W., Pilegaard, K., Rannik, Ü., Rebmann, C., Suyker, A., Tenhunen, J., Tu, K., Verma, S., Vesala, T., Wilson, K., Wofsy, S., 2001. Gap filling strategies for defensible annual sums of net ecosystem exchange. Agricultural and Forest Meteorology 107 (1), 43–69.

Farquhar, G.D., von Caemmerer, S., 1982. Modelling of photosynthetic response to the environment. Physiological Plant Ecology II. Encyclopedia of Plant Physiology, New Series, vol. 12B. Springer, Berlin, pp. 549–587.

Gifford, R.M., 2003. Plant respiration in productivity models: conceptualisation, representation and issues for global terrestrial carbon-cycle research. Functional Plant Biology 30 (2), 171–186.

Goudriaan, J., Van Laar, H., 1994. Modelling potential crop growth processes – textbook with exercises. Current Issues in Production Ecology, vol. 2. Kluwer Academic Publishers, Dordrecht.

Haberl, H., Erb, H.K., Krausmann, F., Gaube, V., Bondeau, A., Plutzar, C., Gingrich, S., Lucht, W., Fischer-Kowalski, M., 2007. Quantifying and mapping the human appropriation of net primary production in Earth's terrestrial ecosystems. Proceedings of the National Academy of Sciences of United States of America 104 (July(31)), 12942–12947.

Halley, R.J., Soffe, R.J. (Eds.), 1988. The Agricultural Notebook, 18th ed. Blackwell, London.

Hirose, T., Werger, M.J.A., 1987. Maximizing daily canopy photosynthesis with respect to the leaf nitrogen allocation pattern in the canopy. Oecologia 72 (4), 520–526.

Houghton, R.A., 1999. The annual net flux of carbon to the atmosphere from changes in land use 1850–1990. Tellus B 51 (2), 298–313.

Hoyaux, J., Moureaux, C., Tourneur, D., Bodson, B., Aubinet, M., 2008. Extrapolating gross primary productivity from leaf to canopy scale in a winter wheat crop. Agricultural and Forest Meteorology 148 (4), 668–679.

Huang, Y., Yu, Y., Zhang, W., Sun, W., Liu, S., Jiang, J., Wu, J., Yu, W., Wang, Y., Yang, Z., 2009. Agro-C: a biogeophysical model for simulating the carbon budget of agroecosystems. Agricultural and Forest Meteorology 149 (1), 106–129.

Jamieson, P.D., Semenov, M.A., Brooking, I.R., Francis, G.S., 1998. Sirius: a mechanistic model of wheat response to environmental variation. European Journal of Agronomy 8 (3–4), 161–179.

Janssens, I.A., Freibauer, A., Ciais, P., Smith, P., Nabuurs, G.-J., Folberth, G., Schlamadinger, B., Hutjes, R.W.A., Ceulemans, R., Schulze, E.-D., Valentini, R., Dolman, A.J., 2003. Europe's terrestrial biosphere absorbs 7 to 12% of European anthropogenic CO_2 emissions. Science 300 (5625), 1538–1542.

Johnson, R.C., Mornhinweg, D.W., Ferris, D.M., Heitholt, J.J., 1987. Leaf photosynthesis and conductance of selected triticum species at different water potentials. Plant Physiology 83 (4), 1014–1017.

Jones, H.G., 1992. Plants and Microclimate. Cambridge University Press, Cambridge.

Keating, B.A., Carberry, P.S., Hammer, G.L., Probert, M.E., Robertson, M.J., Holzworth, D., Huth, N.I., Hargreaves, J.N.G., Meinke, H., Hochman, Z., McLean, G., Verburg, K., Snow, V., Dimes, J.P., Silburn, M., Wang, E., Brown, S., Bristow, K.L., Asseng, S., Chapman, S., McCown, R.L., Freebairn, D.M., Smith, C.J., 2003. An overview of APSIM, a model designed for farming systems simulation. European Journal of Agronomy 18 (3–4), 267–288.

Kucharik, C.J., Twine, T.E., 2007. Residue, respiration, and residuals: evaluation of a dynamic agroecosystem model using eddy flux measurements and biometric data. Agricultural and Forest Meteorology 146 (3–4), 134–158.

Li, L., McMaster, G.S., Yu, Q., Du, J., 2008. Simulating winter wheat development response to temperature: modifying Malo's exponential sine equation. Computers and Electronics in Agriculture 63 (2), 274–281.

Liu, L., Kon, H., Matsuoka, N., Kobayashi, T., 2005. Coordination between stomatal conductance and leaf-specific hydraulic conductance in maize (*Zea mays* L.). Journal of Agricultural Meteorology 61 (3), 143–152.

Lokupitiya, E., Denning, S., Paustian, K., Baker, I., Schaefer, K., Verma, S., Meyers, T., Bernacchi, C., Suyker, A., Fischer, M., 2009. Incorporation of crop phenology in Simple Biosphere Model (SiBcrop) to improve land-atmosphere carbon exchanges from croplands. Biogeosciences Discussions 6 (1), 1903–1944.

Mearns, L.O., Mavromatis, T., Tsvetsinskaya, E., Hays, C., Easterling, W., 1999. Comparative responses of EPIC and CERES crop models to high and low spatial resolution climate change scenarios. Journal of Geophysical Research 104 (D6), 6623–6646.

Monje, O., Bugbee, B., 1998. Adaptation to high CO_2 concentration in an optimal environment: radiation capture, canopy quantum yield and carbon use efficiency. Plant, Cell & Environment 21 (3), 315–324.

Moureaux, C., Debacq, A., Hoyaux, J., Suleau, M., Tourneur, D., Vancutsem, F., Bodson, B., Aubinet, M., 2008. Carbon balance assessment of a Belgian winter wheat crop (*Triticum aestivum* L.). Global Change Biology 14 (6), 1353–1366.

Osborne, T.M., Lawrence, D.M., Challinor, A.J., Slingo, J.M., Wheeler, T.R., 2007. Development and assessment of a coupled crop-climate model. Global Change Biology 13 (1), 169–183.

Penning de Vries, F.W.T., Jansen, D.M., ten Berge, H.F.M., Bakema, A., 1989. Simulation of ecophysiological processes of growth in several annual crops. In: Simulation Monographs, vol. 29. Pudoc, Wageningen.

Porter, J.R., Gawith, M., 1999. Temperatures and the growth and development of wheat: a review. European Journal of Agronomy 10 (1), 23–36.

Porter, J.R., Semenov, M.A., 2005. Crop responses to climatic variation. Philosophical Transactions of the Royal Society B 360 (1463), 2021–2035.

Porter, J.R., Bragg, P.L., Rayner, J.H., Weir, A.H., Landsberg, J.J., 1983. The ARC wheat simulation model-principles and progress. In: Opportunities for Manipulation of Cereal Productivity. No. 7 in British Plant Growth Regulator Group Monograph, pp. 97–108.

Ramankutty, N., Evan, A.T., Monfreda, C., Foley, J.A., 2008. Farming the planet: 1. Geographic distribution of global agricultural lands in the year 2000. Global Biogeochemical Cycles 22 (January), GB1003+.

Reichstein, M., Falge, E., Baldocchi, D., Papale, D., Aubinet, M., Berbigier, P., Bernhofer, C., Buchmann, N., Gilmanov, T., Granier, A., Grünwald, T., Havrankova, K., Ilvesniemi, H., Janous, D., Knohl, A., Laurila, T., Lohila, A., Loustau, D., Matteucci, G., Meyers, T., Miglietta, F., Ourcival, J.-M., Pumpanen, J., Rambal, S., Rotenberg, E., Sanz, M., Tenhunen, J., Seufert, G., Vaccari, F., Vesala, T., Yakir, D., Valentini, R., 2005. On the separation of net ecosystem exchange into assimilation and ecosystem respiration: review and improved algorithm. Global Change Biology 11 (September(9)), 1424–1439.

Sakamoto, T., Yokozawa, M., Toritani, H., Shibayama, M., Ishitsuaka, N., Ohno, H., 2005. A crop phenology detection method using time-series MODIS data. Remote Sensing of Environment 96 (3–4), 366–374.

Slafer, G.A., Rawson, H.M., 1994. Sensitivity of wheat phasic development to major environmental factors: a re-examination of some assumptions made by physiologists and modellers. Australian Journal of Plant Physiology 21 (4), 393–426.

Smith, P., Lanigan, G., Kutsch, W., Buchmann, N., Eugster, W., Aubinet, M., Ceschia, E., Béziat, P., Babu, Y.J., Osborne, B., Moors, E.J., Brut, A., Wattenbach, M., Saunders, M., Jones, M., 2010. Measurements necessary for assessing the net ecosystem carbon budget of croplands. Agriculture, Ecosystems & Environment 139, 302–315.

Stockle, C.O., Martin, S.A., Campbell, G.S., 1994. CropSyst, a cropping systems simulation model: water/nitrogen budgets and crop yield. Agricultural Systems 46 (3), 335–359.

Streck, N.A., Weiss, A., Xue, Q., Baenziger, P.S., 2003. Improving predictions of developmental stages in winter wheat: a modified Wang and Engel model. Agricultural and Forest Meteorology 115 (3), 139–150.

Tambussi, E., Nogués, S., Ferrio, P., Voltas, J., Araus, J., 2005. Does higher yield potential improve barley performance in Mediterranean conditions? A case study. Field Crops Research 91 (2–3), 149–160.

Van Laar, H.H., Goudriaan, J., van Keulen, H., 1997. SUCROS97: simulation of crop growth for potential and water-limited production situations. In: Quantitative Approaches in Systems Analysis. No. 14. Wageningen.

Wang, E., Engel, T., 1998. Simulation of phenological development of wheat crops. Agricultural Systems 58 (1), 1–24.

Wang, Q., Watanabe, M., Ouyang, Z., 2005. Simulation of water and carbon fluxes using BIOME-BGC model over crops in China. Agricultural and Forest Meteorology 131 (3–4), 209–224.
Wang, J., Yu, Q., Lee, X., 2007. Simulation of crop growth and energy and carbon dioxide fluxes at different time steps from hourly to daily. Hydrological Processes 21 (18), 2474–2492.
Waring, R.H., Landsberg, J.J., Williams, M., 1998. Net primary production of forests: a constant fraction of gross primary production? Tree Physiology 18 (2), 129–134.
Wattenbach, M., Vuichard, N., Lehuger, S., Sus, O., Li, L., Leip, A., Tormellieri, E., Kutsch, W.L., Buchmann, N., Eugster, W., Dietiker, D., Aubinet, M., Ceschia, E., Bziat, P., Gruenwald, T., Hastings, A., Gottschalk, P., Osborne, B., Cias, P., Cellier, P., Smith, P., 2010. The carbon balance of European croplands: a cross-site comparison of simulation models. Agriculture, Ecosystems & Environment 139, 419–453.
Williams, J.R., Jones, C.A., Kiniry, J.R., Spanel, D.A., 1989. The EPIC crop growth model. Transactions of the ASAE 32, 497–511.
Williams, M., Rastetter, E.B., Fernandes, D.N., Goulden, M.L., Wofsy, S.C., Shaver, G.R., Melillo, J.M., Munger, J.W., Fan, S.-M., Nadelhoffer, K.J., 1996. Modelling the soil-plant-atmosphere continuum in a quercus-acer stand at Harvard, Forest: the regulation of stomatal conductance by light, nitrogen and soil/plant hydraulic properties. Plant, Cell & Environment 19 (8), 911–927.
Williams, M., Eugster, W., Rastetter, E.B., Mcfadden III, J.P., F. S. C., 2000. The controls on net ecosystem productivity along an Arctic transect: a model comparison with flux measurements. Global Change Biology 6 (Suppl. 1), 116–126.
Williams, M., Law, B.E., Anthoni, P.M., Unsworth, M.H., 2001a. Use of a simulation model and ecosystem flux data to examine carbon-water interactions in ponderosa pine. Tree Physiology 21 (5), 287–298.
Williams, M., Rastetter, E.B., Shaver, G.R., Hobbie, J.E., Carpino, E., Kwiatkowski, B.L., 2001b. Primary production of an Arctic watershed: an uncertainty analysis. Ecological Applications 11 (6), 1800–1816.
Williams, M., Schwarz, P.A., Law, B.E., Irvine, J., 2005. An improved analysis of forest carbon dynamics using data assimilation. Global Change Biology 11 (1), 89–105.
Williams, M., Richardson, A.D., Reichstein, M., Stoy, P.C., Peylin, P., Verbeeck, H., Carvalhais, N., Jung, M., Hollinger, D.Y., Kattge, J., Leuning, R., Luo, Y., Tomelleri, E., Trudinger, C., Wang, Y.-P., 2009. Improving land surface models with FLUXNET data. Biogeosciences Discussions 6 (2), 2785–2835.
Wullschleger, S.D., 1993. Biochemical limitations to carbon assimilation in C_3 plants – a retrospective analysis of the A/Ci curves from 109 species. Journal of Experimental Botany 44 (May(5)), 907–920.
Xue, Q., Weiss, A., Baenziger, P., 2004. Predicting phenological development in winter wheat. Climate Research 25 (3), 243–252.
Yan, W., Wallace, D.H., 1998. Simulation and prediction of plant phenology for five crops based on photoperiod × temperature interaction. Annals of Botany 81 (6), 705–716.
Yang, H.S., Dobermann, A., Lindquist, J.L., Walters, D.T., Arkebauer, T.J., Cassman, K.G., 2004. Hybrid-maize – a maize simulation model that combines two crop modeling approaches. Field Crops Research 87 (2–3), 131–154.
Ye, Z.-P., Yu, Q., 2008. A coupled model of stomatal conductance and photosynthesis for winter wheat. Photosynthetica 46 (4), 637–640.